

# Improving India's National Household Income Survey

*An examination of survey design and field implementation issues for the National Household Income Survey 2026*

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Masters of Development Practice, 2024-2026

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## Executive Summary

India is preparing to undertake the mammoth task of conducting a National Household Income Survey (NHIS) in 2026, the country's first dedicated income survey administered by the official statistical system. The survey will cover approximately 4,50,000 households across rural and urban India and is the most ambitious distributional measurement exercise the country has attempted. With existing consumption-based estimates increasingly criticized for understating the top of the distribution, and India's economic structure shifting away from agriculture toward services and self-employment, the integrity of this survey carries direct implications for tax policy, welfare programme targeting, and the credibility of national inequality estimates.

India has previously attempted to collect income data through the National Sample Survey on four separate occasions between 1955 and 1984, and each attempt was discontinued due to high non-response and systematic underreporting. The NHIS 2026 represents the first attempt at collecting income specifically by the official statistical system to build on these methodological foundations and produce a permanent, publicly accessible income series. The challenge is that income measurement in India does not align cleanly with the assumptions embedded in standard income surveys, and the design choices made in the schedule will shape the quality of every downstream estimate.

This paper evaluates the NHIS 2026 schedule against ten interlocking issues, organized into survey design issues and field implementation issues. For the non-agricultural self-employment module (Block 7c), the analysis is quantitative. A synthetic ground-truth population of 75,000 households calibrated to PLFS 2024-25 is constructed, and the NHIS measurement rule is applied to it directly under two passes. For the remaining issues, including migration-linked income, imputed rent, field visit timing, enumerator gender dynamics, module sequencing, enumerator training, and the capture of top incomes, the analysis is comparative and descriptive, drawing on income-survey practice in Bangladesh, the Philippines, Brazil, South Africa, Malaysia, and others, alongside primary interview evidence from practitioners at MOSPI, IDinsight, ASER (Pratham India), ResGov and the Civil Services (Economics, Revenue and Audit).

The synthetic data exercise produces a clear finding. Under design-only conditions, with every respondent reporting their true last-month revenue, expenses, and months of operation, the NHIS Block 7c module under-estimates non-agricultural self-employment income by approximately 10 per cent at the all-India level. This bias is structural to the estimator and is absorbed largely by the panel rotation design. The headline result, however, is what happens when even a modest field reporting bias is layered on. A conservative five per cent revenue under-report combined with a five per cent expense over-report widens the

under-measurement to nearly 40 per cent. The amplification mechanism is the months-of-operation multiplier, which carries any field-level error through the annualization step at up to twelvefold scale. The implication is that field implementation choices are several times more consequential for the accuracy of Block 7c estimates than the estimator's design itself.

The recommendations that follow are organized into ten items, divided between survey design and implementation. On the design side, the paper recommends retaining the last-month recall window in Block 7c given the 1.5 per cent share of seasonal and sequential households, treating the 30 percentage point reporting-driven gap as the base case for under-measurement, adding a screening question for migrant household members, ensuring operating costs reflect post-subsidy prices, including imputed rent for urban households only, piloting an embedded split-sample experiment on module sequencing in the next round, and refining three instrument variables to enable downstream integration with administrative tax records. On the implementation side, the paper recommends extending the visit window beyond standard office hours and flagging proxy responses, ensuring each field team includes at least one female enumerator with adjusted entry requirements, and redesigning enumerator training around realistic situational scenarios with explicit confidentiality framing.

With the NHIS 2026 schedule already finalized and fieldwork underway, the recommendations in this paper are positioned to inform the next round of the survey rather than the current one. The World Bank Global Poverty and Inequality Data Team intends to take these recommendations forward to MOSPI directly, as part of the Bank's continuing technical engagement on India's distributional measurement infrastructure. On the implementation side, training and visit-timing recommendations can be adopted earlier through field protocol updates without requiring a redesign of the schedule itself. The embedded split-sample experiment on module sequencing should be planned now so that it can be incorporated into the next survey's design rather than retrofitted after fielding begins. The findings of this paper are intended to support the NHIS in delivering on what it has been launched to do and produce a credible measurement of how income is earned, distributed, and unequally held across Indian households.

# 1. Problem Statement

## 1.1 Introduction

India is preparing to undertake the mammoth task of conducting a national household income survey. This is an exercise historically undertaken mostly by higher-income economies, while India and other low and middle-income countries have relied on consumption expenditure surveys to calibrate poverty and inequality measures (Lakner et al., 2016). The move marks a meaningful shift in the country's statistical infrastructure, and two factors are driving it. First, India's economy is gradually moving toward a more service-based composition, with the share of the workforce engaged in services rising and that engaged in agriculture declining steadily over the last two decades (MOSPI, 2024b). Second, India's existing consumption-based inequality estimates have come under increasing scrutiny for understating the distribution at the top, and a dedicated income survey is seen as the instrument best placed to address this gap (Shukla, 2025).

The reason why this divide in usage of an income or a consumption expenditure survey exists comes down to the structure of the economy. Deaton (1997) through his study reasons how in low and middle-income settings, consumption was considered a more reliable welfare measure than income. For economies where the majority of the population is engaged in agriculture or informal self-employment, income becomes difficult to measure, especially in cases where the economic activity comprises goods for own consumption. In addition, these sources of income tend to fluctuate significantly over short periods, while consumption typically smooths over time. Richer, more industrialized economies face the opposite problem and have therefore tended to rely on income surveys, which use less intensive instruments and shorter fieldwork.

Consumption and income capture different notions of welfare, and the choice between them materially moves the measured inequality even when the underlying distribution is unchanged. Consumption reflects the level of welfare a household actually realizes, while income reflects the level it could have a possible command over. For poorer households the distinction is small, since most income is consumed. For richer households the gap widens, as a larger share of income is saved rather than spent (Lakner et al., 2016). The choice of welfare aggregate therefore does not move the poverty rate much, since poverty focuses on the bottom of the distribution, but it does move the inequality measure where the entire distribution counts. Haddad et al. (2024) find that Gini coefficients from consumption surveys are on average 4.5 points lower than those from income surveys for the same country.

Empirical work on India confirms this gap directly and shows that it is large enough to change how the country is positioned globally. Using the India Human Development Survey, which collects both consumption and income data for the same households, Himanshu (2019) estimates an income Gini of 0.54 in 2011-12 against a substantially lower consumption Gini, with the income figure placing India among the more unequal countries globally and the consumption figure placing it among the more equal ones. Shukla (2025) extends this to 2022-23 and reports a consumption Gini of 0.357, an income Gini of 0.410, and a saving Gini of 0.560. The consumption number alone tells one story, the income and saving numbers tell another, and the gap between them is structural. Rankings in global comparisons may therefore reflect the choice of survey instruments as much as real differences in inequality (Lakner et al., 2016), and comparisons between countries whose estimates are based on different survey types are not advisable.

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**Box 1.1: What India's "fourth most equal country" ranking does and does not show**

In 2025, the Press Information Bureau cited World Bank data placing India among the four most equal countries in the world, alongside Norway, Belgium, and Finland (Press Information Bureau, 2025). The figure draws on the Household Consumption Expenditure Survey 2022-23, and a World Bank technical note has cautioned that the estimate may understate inequality on account of sampling and data limitations (World Bank, 2025). The ranking has been read as evidence that India is one of the world's most equal economies, which is misleading given that comparisons between countries using different welfare aggregates and survey methods are not advisable, as established earlier in this section. What it actually reflects is a combination of survey changes and methodological choices in the underlying data, which together pull the headline number well below what a like-for-like comparison would produce.

The largest of these changes sits in the consumption aggregate itself. Himanshu (2026) shows that the World Bank's adjusted Gini for India is roughly four percentage points lower than the unadjusted Gini in both 2011-12 and 2022-23, a magnitude exceeding what most countries see in an entire decade. Three adjustments drive most of this. The World Bank now adds the imputed value of free transfers, food, clothing, and footwear to lower-income households, raising the bottom of the distribution. It excludes durables, jewellery, watches, and furniture, which are concentrated at the top. It also excludes institutional health expenditure. Each of these choices is defensible in isolation. Together they compress the measured Gini in a way that has no equivalent in the income-based estimates used by most high-ranked countries, including Norway, Belgium, and Finland.

Recall period changes also compound this. The 2022-23 HCES uses the Modified Mixed Reference Period, which the World Bank (2018) documented produces materially different headline numbers than the Uniform Reference Period used in earlier rounds, with India's 2011-12 extreme poverty rate falling from 22.9 to 13.4 per cent under MMRP for reasons driven by questionnaire structure rather than by any change in actual consumption (World Bank, 2018; World Bank, 2024). The apparent decline in inequality between earlier rounds and 2022-23 cannot therefore be cleanly separated from the change in measurement. The ranking is best read as a statement about what the 2022-23 HCES is set up to capture, not as evidence of where India sits on the global inequality distribution. Resolving this requires income data to support the consumption expenditure data, which the NHIS 2026 is positioned to provide, and a wider data infrastructure that supports cross-source validation rather than reliance on any single instrument (World Bank, 2021).

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In 'Accelerating India's Development', Karthik Muralidharan (2024) draws a useful analogy, noting that policymaking without timely and credible data is akin to flying a modern aircraft without real-time information. The point translates directly to inequality measurement. Without a clear picture of how income is distributed, the state ends up making decisions about redistribution, taxation, and welfare targeting based on assumptions rather than evidence. For a country of India's scale and diversity, the cost of getting this wrong is significant. Strengthening the underlying data systems is therefore central to improving state capacity, and ensuring that this data remains robust is one of the most pressing needs of the moment.

## 1.2 Why India Needs a Household Income Survey

The case for India to invest in a dedicated income survey rests on several interlocking reasons, of which accurate inequality measurement is one. Income data also enables a clearer reading of how households actually sustain their living standards, how savings are distributed, how income is earned across labor, capital, and self-employment, and how fiscal policy can be designed and evaluated. Each of these is poorly served by consumption data alone.

- An income survey complements a consumption survey by clarifying the sources that sustain household living standards. Consumption data shows what households are able to spend, but income data reveals whether that spending is supported by earnings, savings, transfers, or credit. This distinction is particularly important for low and lower-middle income households, where consumption smoothing can conceal

underlying income fragility. In India, labour market conditions are often tracked through fragmented surveys and proxy indicators. Credible income data can therefore improve how consumption patterns are interpreted and strengthen the evidence base for labour and employment policy. It helps show whether households are advancing through stronger earnings or simply coping to maintain living standards (Deaton, 1997).

- Tax progressivity and fiscal policy design depend on a clear reading of the underlying income distribution. Evaluating whether a tax system is progressive requires consistent measurement of pre-tax and post-tax incomes (Saez & Zucman, 2023). Optimal taxation frameworks further show that the design of progressive tax systems depends critically on the structure and extent of inequality in the economy (Heathcote et al., 2017). Higher income inequality is also associated with worse social outcomes, including lower life expectancy, higher prevalence of mental illness, and greater incidence of chronic disease (Wilkinson & Pickett, 2006, 2009).
- Understanding the composition of income, that is, how much comes from wages, capital, business, and transfers, matters for designing fiscal policy. A consumption survey records what households spend but cannot tell whether that spending is funded by salaried earnings, self-employment profits, returns on capital, or government transfers. The distinction is consequential, since each source faces different tax treatment, different exposure to economic shocks, and different policy levers. Without source-level income data, tax incidence analysis, subsidy targeting, and the evaluation of programmes such as direct benefit transfers operate on assumption rather than evidence (Shukla, 2025).
- For low and lower-middle income countries, rising inequalities could point towards issues in public delivery systems as well as systemic issues that contribute to the problem. Their identification is implausible until countries are able to regularly and accurately measure their inequality levels (López-Calva, Filmer, & Fu, 2025).
- Whether growth is broadly shared or concentrated at the top has important consequences both for economic performance and for how individuals experience opportunity and exclusion. Evidence from both macro and micro settings shows that higher inequality can weaken growth, particularly when it reflects a widening gap between low-income households and the rest of the population, limiting human capital formation and reducing the ability of disadvantaged groups to participate in and benefit from growth (Ostry et al., 2014). These inequalities also shape everyday

lived experiences through differences in social interactions, access to networks, and perceptions of opportunity, which in turn influence how individuals experience mobility and exclusion. Payne et al. (2017) further emphasize that policy interventions may fail to gain public support if they do not engage with these lived realities of class and hierarchy.

These arguments only scratch the surface, but they suggest that the case for an income survey in India is built on multiple foundations rather than on inequality measurement alone. Each of these pillars points in the same direction. Consumption data, while central to poverty measurement, is increasingly insufficient for capturing the full picture of how Indian households earn, save, and absorb risk (Lakner et al., 2016). Expanding data infrastructure to collect household income data is therefore necessary to complement existing consumption-based measures and improve the measurement of distributional outcomes.

At the same time, we also know that income measurement does not align easily with the economic structure of developing countries such as India. High informality, irregular earnings, and weak administrative records pose significant challenges. The following section examines these constraints in detail.

### 1.3 Issues Pertaining to Income Measurement in India

India has had a long history of trial and error with collecting reliable income data. The Press Information Bureau (2025) documents that multiple attempts between 1955 and 1984 to introduce income-based household surveys were eventually discontinued due to persistently high non-response and systematic underreporting, particularly among higher-income households. The most concrete diagnosis came from the rounds in the late 1960s, when income reported by households was found to be lower than reported consumption and savings combined, an internal inconsistency that no methodological revision could resolve (Shukla, 2010). Because income modules were integrated within consumption surveys, they were also found to adversely affect response quality across both income and consumption questions (Deaton, 1997). The scale of these issues limited the viability of such surveys and raised concerns about their ability to generate credible national estimates, leading to a continued reliance on consumption-based surveys for welfare measurement.

The gap left by these failed attempts was partially filled by private and academic efforts. NCAER's National Survey of Household Income and Expenditure, conducted in 2004-05, was the first dedicated income survey in India to follow international methodological standards and demonstrated that direct income measurement is feasible at scale (Shukla, 2010). The People Research on India's Consumer Economy ran the ICE 360° surveys in 2014,

2016, and 2021, building on NCAER's framework with explicit stratification across income groups. These efforts established the methodological foundation for direct income measurement in the Indian context, but they were never integrated into the official statistical apparatus, and their findings therefore had limited influence on national policy discourse.

These challenges highlighted about failed previous attempts reflect deeper structural and operational constraints, including informality, seasonal income variation, and limitations in survey design and implementation. Any renewed effort to implement an income survey in India will need to address these issues systematically and upfront to ensure credible data collection. The following subsections examine these challenges in detail.

**Table 1: Summary of Issues Pertaining to Income Measurement in India**

| S. no.                      | Issue                                                                                                                                              | Severity         | Population segment most affected                                                        | Existing survey design                                                                                                                                                                                           |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Survey Design Issues</b> |                                                                                                                                                    |                  |                                                                                         |                                                                                                                                                                                                                  |
| <b>1</b>                    | Last-month recall in the non-agricultural self-employment module amplifies field-level reporting error through the months-of-operation multiplier. | High             | Non-agricultural self-employed households (around 13 to 18 per cent of all households). | Panel rotation across April, July, October, and January absorbs most seasonal variation, leaving a residual design-only bias of around 10 per cent that widens to 40 per cent under a 5 per cent reporting bias. |
| <b>2</b>                    | Migration-linked income is captured only as remittances, not as total earnings of the absent member.                                               | Moderate         | Migrant-sending households, around 5 to 18 per cent of households depending on state.   | The schedule has no screening question for absent earners; remittances are folded into generic transfer income.                                                                                                  |
| <b>3</b>                    | Imputed rent is excluded, departing from international standards and compressing the upper tail of the income distribution.                        | Moderate to High | Owner-occupier households, particularly in urban areas.                                 | Excluded on grounds of estimation difficulty in thin rural rental markets, but with no separate treatment for urban areas where rental markets are active.                                                       |

| S. no.                       | Issue                                                                                                                                                                        | Severity         | Population segment most affected                                                                            | Existing survey design                                                                                                                                                                                         |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4                            | Top of the income distribution cannot be reached through field protocol alone, and the schedule does not collect the variables needed for downstream tax-record integration. | High             | Higher-income households, particularly those in gated communities.                                          | Pre-testing acknowledges access barriers. Matching variables (granular industry codes, legal form of enterprise, occupation classification aligned with ITR categories) are absent or insufficiently granular. |
| 5                            | Treatment of subsidies and in-kind transfers in the operating-cost and income aggregates is not made explicit in the design guidance.                                        | Moderate         | Self-employed and agricultural households, low-income households receiving in-kind transfers.               | Expenditure is captured at the point of transaction and in-kind transfers are recorded under their own line items in Block 8, but the design guidance does not flag this as a deliberate choice.               |
| <b>Implementation Issues</b> |                                                                                                                                                                              |                  |                                                                                                             |                                                                                                                                                                                                                |
| 6                            | Visit timing affects whether the primary earner is reached, and proxy reporting introduces avoidable measurement error on the income module.                                 | High             | Households where the primary earner is absent during daytime visits, dual-earner and gig-worker households. | Current MoSPI practice follows standard 9 AM to 5 PM hours with limited flexibility. Pre-testing recommends scheduling flexibility but the protocol has not been updated.                                      |
| 7                            | Enumerator-respondent gender dynamics affect participation and response quality on sensitive modules.                                                                        | Moderate to High | Women respondents and households in conservative settings.                                                  | Pre-testing recommends gender-balanced teams. Women remain underrepresented in MoSPI's selection process due to entry requirements and safety and mobility concerns.                                           |
| 8                            | Module sequencing places income before expenditure, which anchors expenditure responses and weakens                                                                          | Moderate         | All surveyed households, particularly those where income and expenditure are reported by different members. | Current sequence follows NSS convention. The instrument does not include a live consistency check between the two modules.                                                                                     |

| S. no. | Issue                                                                                                                                                                                 | Severity | Population segment most affected                                                                   | Existing survey design                                                                                                                                                                       |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        | any downstream consistency check.                                                                                                                                                     |          |                                                                                                    |                                                                                                                                                                                              |
| 9      | Enumerator training is theoretical rather than scenario-based and does not prepare enumerators for income-specific situations such as refusal, tax anxiety, or implausible reporting. | High     | All surveyed households, with the strongest effect on higher-income and complex-income households. | Pre-testing recommends focused training designed for personnel undertaking the data collection work, but the current training duration and format are not yet designed for an income survey. |
| 10     | Long interview duration creates respondent fatigue and satisficing across the latter modules.                                                                                         | Moderate | All surveyed households, particularly those with multiple income sources.                          | Pre-testing shows the average interview lasts around 100 minutes, beyond the typical reliable response window of 90 minutes for financial-information modules.                               |

## Survey Design Issues

### 1.3.1 Measuring self-employment income under seasonality and informality

Measuring income from self-employment activity is structurally hard because seasonality enters through both the operation calendar and the revenue and expenditure patterns within an operating month. According to PLFS 2023-24, 58.4 per cent of workers are self-employed (MOSPI, 2024b), and Deaton (1997) notes that for these households, income is not directly observed but constructed from revenues and costs, with profits often hard to separate from household consumption. Near subsistence, this produces reported incomes that appear close to zero despite ongoing economic activity. The further complication is that revenue and expenditure within a self-employed household do not move in lockstep across the year. Construction households face monsoon troughs, retail and personal services peak around the festival season, and education-related work follows the academic calendar. The choice of recall period therefore involves a structural trade-off. Shorter recall windows tend to improve reporting accuracy because respondents remember recent transactions more reliably, but they risk capturing only a partial slice of an annual income that varies substantially across months. Longer recall windows capture seasonal patterns but introduce memory decay and a tendency to estimate rather than report (World Bank, 2018).

A further issue concerns how operating costs are valued. As Himanshu (2026) notes, consumption surveys typically value goods at market prices and do not net out subsidies. Extending this approach to the operating-cost modules of an income survey risks overstating costs and understating real purchasing power, particularly for poorer households whose inputs come through subsidized channels. Operating costs in self-employment and agricultural modules should therefore reflect actual prices paid after subsidies. In-kind transfers raise a related but distinct issue. They affect welfare but do not represent liquid purchasing power and folding them into the cash income aggregate distorts measured inequality. The cleaner treatment is to record them separately, which preserves their welfare relevance without contaminating the income measure. This is something that the current NHIS questionnaire does well.

We use a synthetic-data analysis in Section 3.1.1 to examine this problem directly and see whether the NHIS Block 7c module, which uses a last-month recall window for non-agricultural self-employment, produces a structurally biased annual income estimate when applied to a population that exhibits these seasonality patterns, and how a layer of plausible field reporting bias propagates through the estimator.

### *1.3.2 Migration-Linked Income*

One part of the survey design that becomes difficult to ignore on closer inspection is how migration is treated. The National Household Income Survey module does not explicitly provide a way to identify whether any member of the household is currently living elsewhere for work. This absence creates a meaningful gap in what the survey is able to capture.

Migration in India is not peripheral. As of 2020-21, 11.4 per cent of men were migrants, and 48.8 per cent of male migrants had moved for employment-related reasons (Singh, 2024). This matters for income measurement because, in such households, earnings are often generated outside the household but realized within it through remittances. Earlier work has similarly shown that migration in India often produces "missing men" in source regions, making migration central rather than incidental to household livelihood strategies (Tumbe, 2014). This is reinforced by interview evidence from Rohan Raj at IDinsight, who noted that, in a 2025 survey representative of 12 states, 18 per cent of households had one or more migrant members, with the share rising to 5.9 per cent reporting employment-related migration in Telangana alone, where 50,000 households were surveyed and close to half of those households received remittances. Although remittance incidence was not collected for the full 12-state sample, the broader pattern was reported to be common across states. Together, these findings point to migration as a substantial and policy-relevant source of household income.

Three distinct design ambitions sit underneath this issue. The first is identifying migrant-sending households, which requires a screening question on the roster. The second is measuring remittance receipts, which the current schedule already does but folds into generic transfer income. The third, and the hardest, is estimating total earnings of the migrant member, which requires either a proxy report from a household member who knows the absent earner's income or a separate instrument administered at destination. The current NHIS attempts only the second, and even that without distinguishing remittance income from other transfers. When the survey reaches a migrant-sending household, the respondent is typically someone managing the household in the migrant's absence, and what gets reported is the remittance received, not the total income earned. Remittances are only a fraction of the migrant's earnings, with a portion already consumed at the destination, and treating them as total income effectively compresses the income measure. Conversations with employees at the Ministry of Statistics and Programme Implementation, both of whom have been involved in field data collection, confirm that migrant members are often not explicitly tracked in the survey instrument (MoSPI staff member, interview, 2026).

What this points to is not a failure of coverage but of representation. Migrant households are present in the data, but the way their income is recorded does not reflect how it is actually generated. Other large migrant economies have addressed this at the instrument level, treating migration as a structural design issue rather than as a special case, and the analysis in Section 3.1.2 examines what India can learn from those examples.

### *1.3.3 Imputed Rent*

The Canberra Group Handbook on Household Income Statistics is one of the key references for constructing income surveys and was also used in shaping the current National Household Income Survey of India, as confirmed in an interview with an officer at MoSPI involved in the construction of the National Household Income Survey. The handbook adopts a broad definition of income and includes the net value of housing services from owner-occupied dwellings, commonly referred to as imputed rent. In effect, homeowners are treated as consuming a housing service from their own dwelling, valued as the market rent of a comparable unit net of housing-related costs such as interest, maintenance, and taxes (United Nations, 2011).

The difficulty begins when this needs to be implemented. Estimating imputed rent requires either reliable rental markets or detailed information on housing characteristics and costs. In many parts of India, especially in rural areas, rental markets are limited or do not exist in a meaningful way. In such cases, the imputation becomes heavily dependent on assumptions rather than observed data. Himanshu (2026) pushes this concern further. He

argues that imputed rent is better understood as consumption from an asset rather than current income. When included in income aggregates, particularly in contexts with high homeownership and weak rental markets, it can shift measured income without reflecting actual purchasing power. The problem is not only one of estimation, but of interpretation.

The recommendation that follows from the analysis in Section 3.1.3 is to include imputed rent for urban households and exclude it for rural households, since rental markets in urban India are active enough to support reliable imputation while rural markets are too thin. This rule is defensible on accuracy grounds but carries a comparability cost. Different aggregate definitions across rural and urban areas mean that direct rural-to-urban comparisons of measured income require an explicit adjustment, and any state-level or national aggregate has to flag the mixed treatment to remain transparent.

#### *1.3.4 Reaching the Top of the Income Distribution*

Higher-income households present a different kind of measurement challenge. Participating in a government-led income survey offers them little benefit while raising real concerns about scrutiny and tax implications, even when the survey is legally confidential and the data are held within the statistical system. The pre-testing report documents both the access and the response problems directly, with gated communities frequently restricting enumerator entry altogether and self-enumeration sometimes proposed as a workaround that does not in practice resolve either issue (MOSPI, 2025; Deepak, interview, 2026). This compounds the broader issue of underreporting at the top of the distribution. Singh (2025) shows that reporting behavior is not random; higher-income and wealthier households tend to report disproportionately lower incomes relative to their true economic position, leading to a compression of the observed income distribution. Disclosure may be associated with taxation, loss of benefits, or social scrutiny, and the very segments that are most important for measuring inequality accurately are also the least likely to report income truthfully.

No realistic adjustment to the field protocol will produce representative coverage at the top of the income distribution. The more productive question, examined in detail in Section 3.2.4, is how to make the survey data useful for characterizing the top through other means, particularly through statistical integration with administrative tax records as has been done by Brazil, South Africa, and Malaysia. This requires the NHIS instrument to collect, at sufficient granularity, the variables that allow downstream researchers to match survey records to tax data, which is a design choice that needs to be made upfront rather than at the analysis stage.

## Implementation Issues

### *1.3.5 Timing of the Field Visit and Proxy Respondents*

Collecting income data accurately depends critically on interviewing the primary earning member of the household. In practice, however, this is rarely guaranteed. Information about income is often not fully shared within households, and non-earning members may only have partial or approximate knowledge. As a result, interviewing a proxy respondent can directly introduce measurement error, particularly for complex or multi-source income streams. Survey teams have limited windows during which households are accessible. Mornings are typically rushed, as individuals leave for work, while evenings are less conducive as respondents are tired or unwilling to engage. During the day, the most accessible respondent is often a non-earning household member, usually a woman managing the household. While this works well for collecting expenditure and consumption data, it becomes problematic for income reporting, which is often less visible and less discussed within the household. This issue was also highlighted in conversations with a MoSPI survey team member involved in HCES and CPI data collection (MoSPI field staff, interview, 2026).

### *1.3.6 Enumerator-Respondent Gender Dynamics*

Another constraint emerges from enumerator-respondent dynamics. In many parts of India, women may be uncomfortable interacting with male enumerators or allowing them into the household. Evidence from survey practice shows that enumerator gender can significantly affect both participation and response quality (Fuletra & Joshi, 2024). This was further confirmed in an interview with Isha Fuletra, who noted that gender matching between enumerators and respondents can improve data reliability. At the same time, ensuring the availability of female enumerators is not straightforward. A recurring recommendation across interviews was the need to lower entry requirements and extend training periods to improve the participation and retention of female survey staff (I. Fuletra, interview, 2026; Deepak, interview, 2026).

### *1.3.7 Module Sequencing*

The order in which modules are administered also affects response quality. The current NHIS schedule places income modules before the consumption expenditure module, consistent with standard NSS convention. The concern is that asking sensitive income questions first allows respondents to anchor their later expenditure answers to what they have already disclosed, which makes any downstream consistency check between the two modules far

less meaningful. Sequencing also interacts with the timing issue described above. A respondent who is available during the daytime is often not the primary earner, but can answer expenditure questions reliably, which means starting the interview with expenditure builds engagement before the harder income modules begin. The analysis in Section 3.2.2 examines this trade-off in more detail, drawing on international examples and interview evidence from practitioners who have run similar surveys.

### *1.3.8 Enumerator Training*

Enumerator training is a separate constraint. Income surveys present scenarios that simply do not arise in consumption surveys, including managing refusal from higher-income households, handling respondent anxiety about tax implications, and recognizing when stated income is implausible given observed assets in the household. Training that is theoretical rather than scenario-based is particularly poorly suited to these situations. Sharad Pandey noted in an interview that government enumerators typically receive minimal training before deployment, and that what training exists tends not to be grounded in field realities (S. Pandey, interview, 2026). The recommendation that emerges in Section 3.2.3 is to redesign training around realistic situational scenarios, with explicit attention to respondent framing and to confidentiality messaging that helps build the trust needed to obtain honest responses.

### *1.3.9 Respondent Fatigue and Interview Length*

Respondent fatigue is a major concern in income surveys. The Brief Report on Pre-testing of Schedule of Household Income Survey shows that the average interview takes around 100 minutes, which is unusually long. As interviews progress, respondents may lose attention, provide approximate answers, or try to finish quickly, weakening data quality (MOSPI, 2025). This concern is reinforced by interview evidence from Rohan Raj, who noted that surveys generally struggle to maintain accurate responses beyond 90 minutes, and that for modules collecting financial information, it is often difficult to retain respondents even for 45 to 60 minutes. Given that income surveys are likely to feel both demanding and sensitive, a 100-minute completion time raises serious concerns about response quality and suggests the need for a shorter and more focused instrument. Alongside this, the pre-testing report shows that even within a 100-minute window, respondents struggle with reference periods, with 47 per cent demonstrating only moderate understanding and 13 per cent showing only little understanding of the time periods they were being asked to report on (MOSPI, 2025). Mahler, Foster, and Tetteh-Baah (2023) underscore this directly, showing that even modest changes in reference-period interpretation can lead to meaningful differences in measured income.

The fatigue and the comprehension problems compound each other. A respondent already 60 or 70 minutes into a sensitive interview is unlikely to ask for clarification on a confusing reference period and more likely to provide an approximate answer that the enumerator will record without flagging.

## 2. Analytical Approach

This paper examines the challenges facing India's National Household Income Survey through two distinct analytical approaches. For the seasonality issue in the non-agricultural self-employment module, where the concern is whether a specific design choice in the instrument introduces structural bias, we construct a synthetic ground-truth population and apply the NHIS Block 7c measurement to it directly. This is described in detail below. For all remaining issues, spanning migration-linked income, imputed rent, field visit timing, gender dynamics, survey training, and the capture of top incomes, the analysis is comparative and descriptive. Each issue is examined through a combination of country-level evidence from income surveys in Bangladesh, the Philippines, Indonesia, Peru, and others, alongside primary interview evidence from practitioners at MoSPI, IDinsight, ASER, and other organizations involved in income data collection in India. Where relevant, the analysis also draws on the NHIS pre-testing report (MoSPI, 2025).

The synthetic data exercise has two direct aims. First, it tests whether the NHIS Block 7c module, which will be administered to approx. 13 percent of rural and 18 percent of urban households engaged in non-agricultural self-employment (Appendix 1), produces a structurally biased estimate of annual income under a plausible data generating process. That is, whether the way the module annualizes last month profit systematically over- or under-estimates true earnings under realistic assumptions about Indian non-agricultural self-employment. Second, it tests how a layer of reporting bias, with respondents under-reporting revenue and over-reporting expenses, propagates through this same estimator to affect the final income estimate.

The exercise cannot validate the true behavioral reporting process. No observational ground truth exists against which to verify the actual rates at which respondents under-report revenue, over-report expenses, or misremember months of operation in the Indian context. The exercise is therefore best read as a structured way of asking how the module performs in measuring annual income for the self-employed non-agricultural population, and how a plausible reporting bias amplifies the estimator's structural weaknesses, rather than as a definitive measurement of NHIS bias in the field.

## 2.1 Synthetic Data Methodology: Non-Agricultural Self-Employment Module

The NHIS 2026 schedule measures non-agricultural self-employment income in Block 7c through a last month recall window. For each business operated by the household (recorded in Q4.5), the enumerator records gross receipts (Q7.9), operating costs (Q7.10), and gross profit or loss (Q7.12) for the last calendar month only. Annual income is then derived analytically by multiplying reported monthly profit by the number of months the business operated in the past 365 days, also recorded in Q4.5. To address the concern that a single month's profit may not be representative of the year, the NHIS design rotates households across four panels surveyed in April, July, October, and January (assuming survey goes on ground in April). The design assumption is that panel-level seasonal variation will average out at the population level.

Since NHIS 2026 is the first income survey of this kind fielded in India, no observational ground truth exists. We therefore construct one synthetically. This is done to test if the last month recall period works well to generate an accurate representation of the incomes earned by households.

The approach is to build a synthetic ground-truth population calibrated to PLFS 2025, apply the NHIS measurement rule to it as written, and compare the resulting estimates to the known truth. The approach is to build a synthetic ground-truth population calibrated to PLFS 2025, apply the NHIS measurement rule to it as written, and compare the resulting estimates to the known truth. The simulation is run in two passes, which serve distinct analytical purposes. Pass A asks whether the estimator is biased even under ideal field conditions. Every respondent reports their true last month revenue, true last month expenses, and true months of operation count. No reporting error is introduced. Any bias that emerges is purely structural, arising from the design of the estimator itself. Pass B asks how much worse outcomes become under plausible field behaviour. Each respondent reports last month revenue at 95 percent of its true value and last month expenses at 105 percent, representing a conservative five percent under-reporting of revenue and five percent over-reporting of expenses. These rates are taken as conservative assumptions consistent with patterns documented in the NHIS pre-test report (MoSPI, 2025). Both passes use the same ground-truth ledger and the same panel assignment, so their difference isolates the contribution of the reporting bias layer.

The synthetic data followed the following design elements:

### *1. Population construction:*

Using PLFS 2025, we approximate that 13 percent of rural and 18 percent of urban households were engaged in non-agricultural self-employment (Appendix 1). On this basis, we construct a sample of 75,000 households containing approximately

100,000 non-agricultural businesses, allocated across 20 major states proportional to each state's share of the non-agricultural self-employed household universe.

The sample size is calibrated as follows. The full NHIS 2026 sample is approximately 4,50,000 households nationally (@GoIStats, 2025). Applying the 13 percent rural and 18 percent urban prevalence rates documented in Appendix 1 to this universe yields a non-agricultural self-employed sub-sample of roughly 65,000 to 72,000 households. We use 75,000 to sit modestly above the upper end of this expected range, providing headroom for state by sector breakdowns and ensuring adequate precision in smaller states. The simulation has been tested under three alternative random seeds and headline bias estimates vary by less than 0.5 percentage points across these runs, indicating that the results are stable to stochastic variation and that 75,000 households is sufficient for the bias estimates we report.

The PLFS 2025 Annual Report tables which were used to inform the structure were as follows:

- a. Table 2 for state household population.
- b. Tables 3 and 4 for state-level prevalence of non-agricultural self-employment for rural and urban sectors respectively.
- c. Table 27 for state-level industry mix, mapped to eleven aggregated categories.
- d. Table 40 for state-level mean monthly self-employment earnings. The realised mean household profit in the simulation matches Table 40 within plus or minus 10 percent for all major states.

Industry sub-category splits within PLFS Table 27 sections (specifically the light versus heavy manufacturing split within Section C, and the personal services versus other services split within Section S) were calibrated against PLFS Table 26, which provides all-India worker distribution at the two digit NIC level. This yields a 58/42 light to heavy manufacturing split and an 82/18 personal to other services split. Appendix 2 documents these derivations in full.

Each business is assigned a twelve month ledger of revenue and expenses spanning March 2026 through February 2027, with an operational or closed flag for each month. Monthly revenue is drawn from a lognormal distribution body, with a Pareto regime imposed on draws above Rs. 5 lakh per month. Chancel and Piketty (2019), in the foundational study of long-run Indian income inequality, treat the upper tail of the distribution as following a Pareto regime where reliable estimation requires data above the 94th percentile of the income distribution. We adopt a substantially more conservative threshold than this literature anchor, set at Rs. 5 lakh per month, which

captures only 0.4 percent of businesses in our simulation. This conservative choice is intentional. Our purpose is not to estimate the upper tail of the income distribution but to evaluate seasonal mismatch in the Block 7c estimator, which operates on the body of the distribution rather than its tail. Pareto households face a design-only bias of approximately –6 percent, slightly less than the –10 percent faced by non-Pareto households, indicating that our headline bias estimates are not driven by the Pareto threshold choice.

## 2. Household typology:

The simulation assigns each household to one of five structural categories reflecting distinct patterns of non-agricultural self-employment in India's informal economy. This matters because the bias introduced by the NHIS last-month estimator differs fundamentally across household types. A simulation that treats all households as structurally similar would misstate both the nature and the distribution of the problem.

**Table 2:** Five categories and their shares in the synthetic population.

| Category | Description                                                                    | Share | Primary bias mechanism                                                                                |
|----------|--------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------|
| 1        | Single or multi-business, continuous year-round, flat revenue and expenses     | ~14%  | Minimal or none as reference month is representative of annual mean                                   |
| 2        | Single or multi-business, continuous year-round, seasonal revenue and expenses | ~76%  | Within-month seasonal mismatch, partially absorbed by panel rotation                                  |
| 3        | Single business, partial-year operation, contiguous operating window           | 1.0%  | Reference month may fall in dark period. NHIS records zero against true positive income               |
| 4        | Multiple businesses, concurrent, year-round                                    | 8.5%  | Same as Cat 1/2. Perennial type, behaves like the continuous group                                    |
| 5        | Multiple businesses, sequential non-overlapping windows                        | 0.5%  | One or both businesses dark in reference month. Structural zeros multiplied by partial active periods |

### *Detailed Classification of Household Categories*

Categories (cat) 1, 2, and 4 together account for approximately 98 percent of the synthetic population, consistent with the NSS 73rd Round finding that 98.3 percent

of unincorporated non-agricultural enterprises in India are perennial (MoSPI, 2016). Category 4, multi-business concurrent households, also belongs in this perennial group because all its businesses operate year-round. Its inclusion as a separate category is motivated by ASUSE 2022-23 evidence that 86.5 percent of unincorporated non-agricultural enterprises are sole Own Account Enterprises (MoSPI, 2024b; Dvara Research, 2026) and the IHDS finding that roughly five percent of urban workers report multiple concurrent activities (Desai et al., 2010), making multi-enterprise households a small but identifiable minority. The number of businesses per household is set at {1: 0.91, 2: 0.08, 3: 0.01}, consistent with these sources.

Within the perennial group, the split between flat (Cat 1) and seasonal (Cat 2) is not set directly but emerges from the industry distribution calibrated to PLFS Table 27 and each industry's seasonality coefficient of variation. Industries with low intra-year variation (heavy manufacturing, professional services, other services) produce Cat 1 households; industries with substantial variation (retail, food services, personal services, light manufacturing, construction, transport, health, and education) produce Cat 2 households. The resulting shares are approximately 14 percent Cat 1 and 76 percent Cat 2, reflecting that most perennial Indian non-agricultural enterprises operate in industries with meaningful seasonal patterns.

Categories 3 and 5, together accounting for approximately 1.5 percent of the population, are the partial-year types and the categories most vulnerable to last-month annualizations. For Cat 3, a reference month falling in the dark period records zero revenue, which the estimator multiplies by the active months-of-operation count to produce a zero annual estimate for a household with substantial true income. For Cat 5, a reference month may catch one business entirely between its two seasonal windows, producing a structural zero that the estimator cannot distinguish from genuine zero income. These mechanisms, rather than seasonal variation within an operating month, are the primary source of estimator vulnerability for these households, and cannot be addressed by panel rotation.

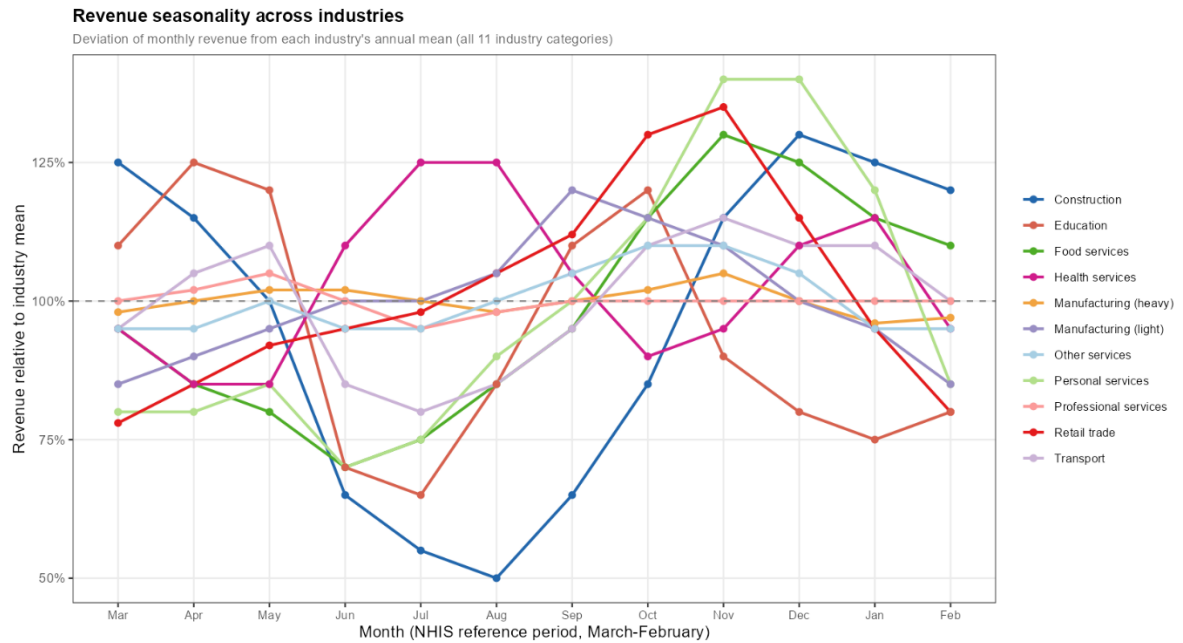
Category 3 covers households running a single business for a contiguous block of five to eight months. Examples include farmers running a provisions store during the agricultural off-season, construction contractors inactive during the monsoon, and private tutors working only during the school year. Category 5 covers households running two non-agricultural businesses in non-overlapping windows of three to five months each, with at least two months between them. This sequential pattern represents activity-switching documented qualitatively by Harriss-White and Gooptu

(2001), such as urban vendors selling firecrackers during Diwali and colours during Holi, or umbrella repairers during the monsoon and fan repairers during summer.

### *3. Seasonality calibration:*

Industry-specific seasonality multipliers are applied to monthly revenue across eleven industry categories. Retail and personal services peak in October and November around Diwali. Food services peak across the wedding season from October through February. Construction troughs sharply during the monsoon months of June through August, falling to approximately 50 percent of the annual mean in August. Education troughs during summer school holidays in June and July and during post-board-exam months in January and February, peaking during exam preparation periods. Health services peak counter-cyclically during the monsoon (waterborne and respiratory illness) and again in winter. Transport, light manufacturing, and other services retain modest intra-year patterns. Heavy manufacturing and professional services are essentially flat across the year. The numerical multipliers used to operationalise these patterns are stylised approximations based on widely understood industry seasonality, applied within the time constraints of this project. A deeper calibration using the PLFS unit-level microdata, which captures earnings across all calendar months through its continuous fieldwork design, would allow these multipliers to be estimated rather than assumed and is a recommended direction for future work. The full set of monthly multipliers by industry is provided in Appendix 2 to make the current assumptions auditable. Figure 1 visualises these patterns.

**Figure 1: Revenue seasonality across 11 industries**



These multipliers are applied uniformly across states, as PLFS does not publish state-level seasonal income data at the required granularity. Expense seasonality is modelled as a damped and one month lagged version of revenue seasonality, reflecting that input procurement (inventory build-up before festival peaks, materials purchase before construction season) typically precedes the revenue peak it supports. For Categories 3 and 5, operating windows are pre-assigned as contiguous blocks at generation. For Categories 1, 2, and 4, all twelve months are operational with industry-specific idiosyncratic closure rates (averaging 0.4 to 2.0 months per year, highest for construction and education).

#### 4. Panel structure:

The four panels correspond to survey months of April, July, October, and January, with reference months of March, June, September, and December respectively. Each household is randomly assigned to a panel, which determines the reference month applied in both simulator passes.

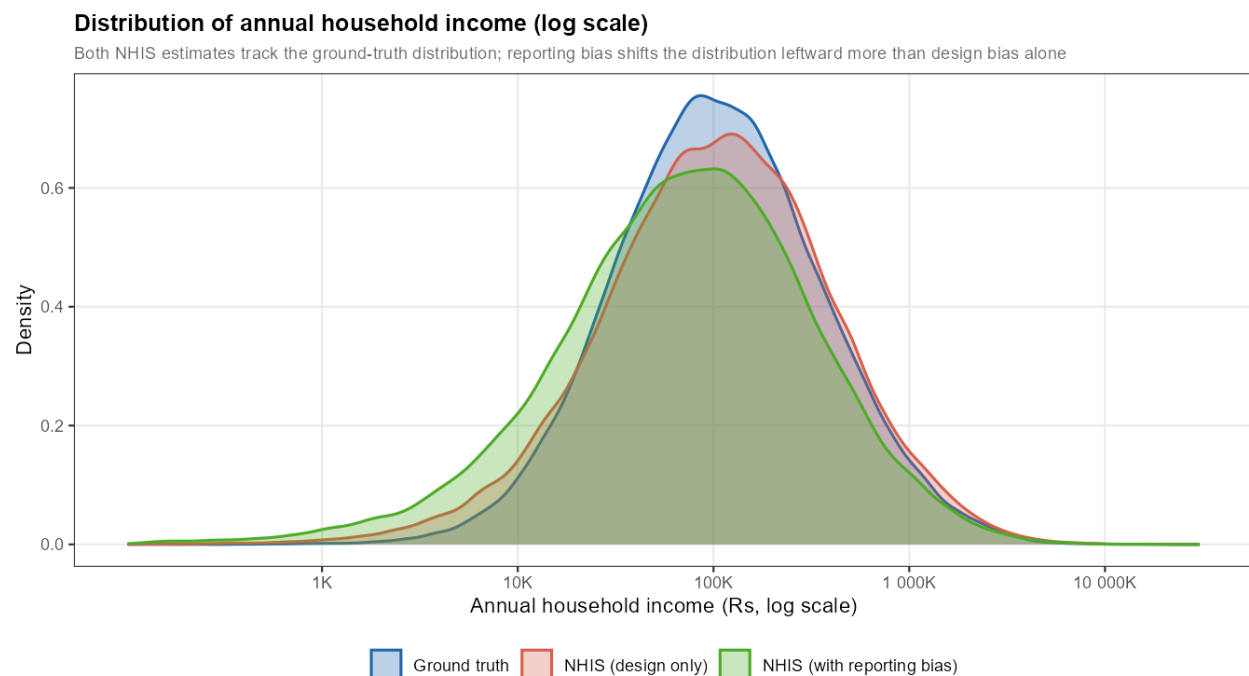
## 3. Analysis

### 3.1 Survey Method Issues

#### 3.1.1 Non-Agricultural Self-Employment Module

Results from the analysis performed using synthetic data are presented through Figure 2 and Table 3. The figure below shows annual household income under three measurements: the simulated truth, the NHIS estimate under design-only conditions, and the NHIS estimate with reporting bias layered on. We elaborate on where these issues originate from, whom they impact the most, and corrective measures in this section.

Figure 2: Distribution of annual household income (log scale)



**Table 3.** All-India NHIS bias under the two analysis passes. A 5% revenue under-report and 5% expense over-report carried through the months-of-operation multiplier amplifies total bias more than four-fold.

| Measure                    | Pass A: Design only | Pass B: Design + reporting |
|----------------------------|---------------------|----------------------------|
| Mean true annual HH profit | Rs. 2,25,995        | Rs. 2,25,995               |
| Mean NHIS-estimated profit | Rs. 2,03,790        | Rs. 1,36,482               |
| All-India bias             | -9.8%               | -39.6%                     |
| Rural bias                 | -10.1%              | -41.8%                     |
| Urban bias                 | -9.7%               | -38.3%                     |

Under pure design, with every respondent reporting their true last month revenue, expenses, and months of operation, NHIS underestimates annual household income from non-agricultural self-employment by 9.8 percent. With a conservative five percent revenue under-reporting and five percent expense over-reporting layered on top, the underestimation widens to 39.6 percent.

The amplification mechanism is the months-of-operation multiplier. A household that under-reports its last month revenue by five percent has that shortfall multiplied by twelve in the NHIS annual estimate. For thin-margin industries the effect is even greater. A construction household with a 15 percent gross margin and a five percent revenue under-report sees its reported monthly profit drop by roughly one-third, sometimes producing a negative annual income estimate for a household with positive true earnings. This means that the quality of field implementation is several times more consequential for the accuracy of Block 7c estimates than the estimator's design itself.

Rural households face slightly deeper bias than urban households across both passes. Rural bias is 10.1 percent under design only and 41.8 percent with reporting, against urban biases of 9.7 percent and 38.3 percent respectively. This modest gap reflects compositional differences in the rural and urban industry mix, with rural areas slightly more weighted toward seasonal industries. The PLFS Annual Report 2025 publishes industry distributions for all workers at the rural and urban level (Table 27) but does not disaggregate these by employment status, so the rural-urban comparison in our calibration relies on the aggregate

worker-level distribution rather than a self-employed-specific one. A more precise decomposition of the rural-urban gap by industry would require analysis of the PLFS unit-level microdata and is recommended for future work. The full set of state-by-sector industry weights, the manufacturing and services sub-section splits, and the seasonality multipliers used in the simulation are documented in Appendix 2 to keep the underlying assumptions auditable for any reader wishing to verify, replicate, or stress-test the calibration.

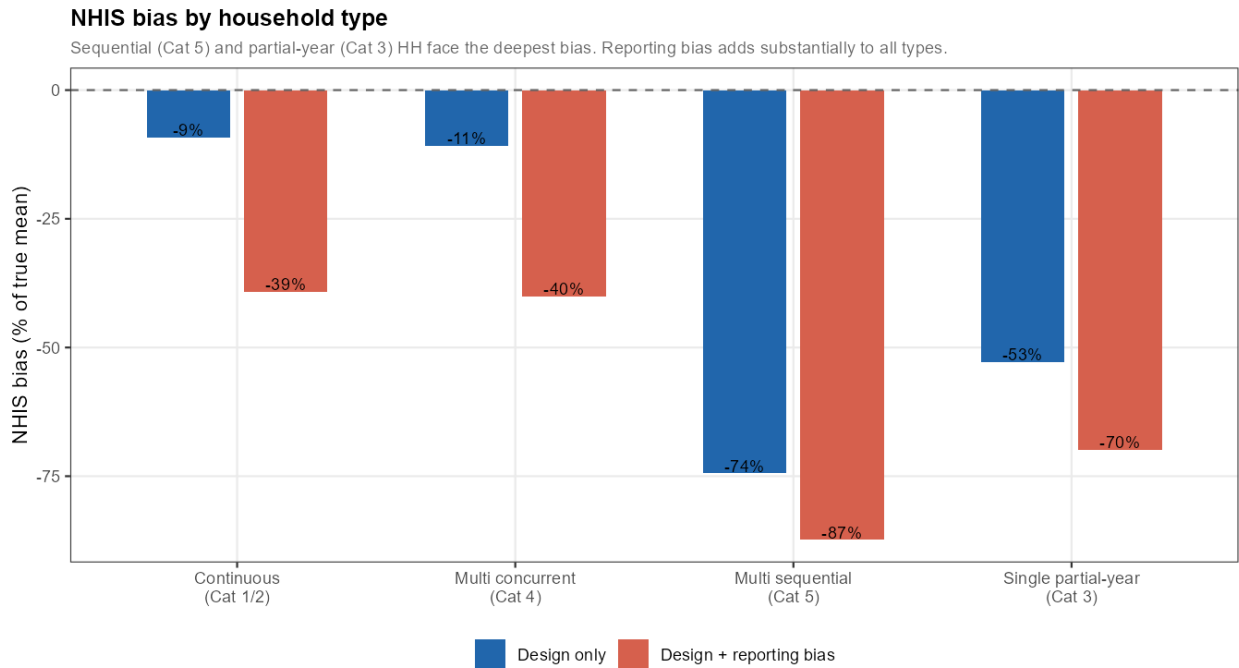
*‘Design only’ bias is concentrated in a minority of households*

The all-India average masks sharp heterogeneity across household types. Table [4] decomposes the bias by category.

**Table 4.** NHIS bias by household type. Cat 4 (perennial concurrent) shows nearly identical bias to Cat 1/2, confirming it belongs in the 98% perennial group. Cat 3 and Cat 5 face structural under-estimation even with no reporting error.

| Household type                       | Share | Design only | Design + reporting |
|--------------------------------------|-------|-------------|--------------------|
| Cat 1/2: Continuous year-round       | 90%   | –9.2%       | –39.2%             |
| Cat 4: Multi-business, concurrent    | 8.5%  | –10.8%      | –40%               |
| Cat 3: Single business, partial-year | 1.0%  | –52.8%      | –69.9%             |
| Cat 5: Multi-business, sequential    | 0.5%  | –74.4%      | –87.4%             |

Figure 3: NHIS bias by household type



From Table 4, we can see how multi-business concurrent households face a design-only bias of 10.8 percent, essentially indistinguishable from the 9.2 percent faced by single-business continuous households, despite running two or three enterprises in parallel. This confirms that Cat 4 is structurally a perennial type and behaves identically under the estimator. The NHIS last-month recall captures a representative operating month for each business, and multiplying by twelve produces a reasonable annual estimate.

Cat 3 and Cat 5 are a different matter. A Cat 3 household whose business operates for five months is surveyed with a probability proportional to its dark period. When the reference month falls in the inactive window, the estimator records zero revenue, multiplies it by the active months-of-operation count, and produces a zero annual estimate for a household with substantial true income. The result is a design-only bias of 52.8 percent. For Cat 5, the mechanism is sharper still. With two non-overlapping windows and gaps between them, a reference month may catch one business between its two seasons entirely, producing structural zeros that the estimator cannot distinguish from genuine zero income. The design-only bias for Cat 5 reaches 74.4 percent, rising to 87.4 percent once reporting bias is layered in.

Together, Cat 3 and Cat 5 account for only 1.5 percent of the synthetic population, consistent with the NSS 73rd Round finding that 98.3 percent of unincorporated non-agricultural enterprises are perennial (MoSPI, 2016). At this share, the structural mismeasurement of partial-year and sequential households does not on its own warrant a fundamental redesign

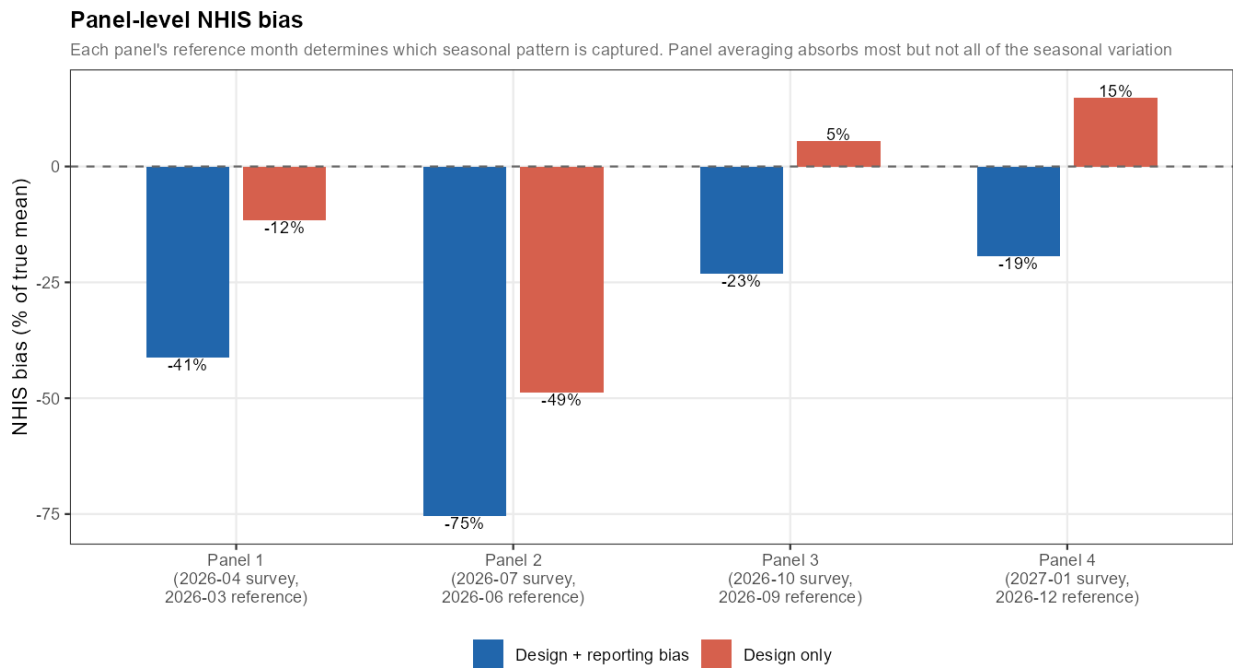
of the Block 7c module. Should subsequent rounds of NHIS, or future ASUSE waves, document a meaningful rise in the share of seasonal or sequential non-agricultural self-employment, the case for adapting the estimator would strengthen materially. The more pressing concern, and the one this analysis returns to, is what happens when even a modest reporting bias compounds the design-only bias. The four-fold amplification documented in Section 4.1 is not specific to the partial-year minority. It applies across the entire universe of Block 7c respondents. The next section examines this layer in more detail.

### *Panel-level variation*

**Table 5.** Panel-level NHIS bias. Panel 2 (June reference) catches the monsoon-season trough across construction, food services, and personal services. Panels 3 and 4 benefit from the festival-season upswing.

| Panel   | Reference month | Design only | Design + reporting |
|---------|-----------------|-------------|--------------------|
| Panel 1 | March 2026      | −11.6%      | −41.3%             |
| Panel 2 | June 2026       | −48.8%      | −75.4%             |
| Panel 3 | September 2026  | +5.4%       | −23.1%             |
| Panel 4 | December 2026   | +14.8%      | −19.4%             |

Figure 4: Panel-level NHIS bias

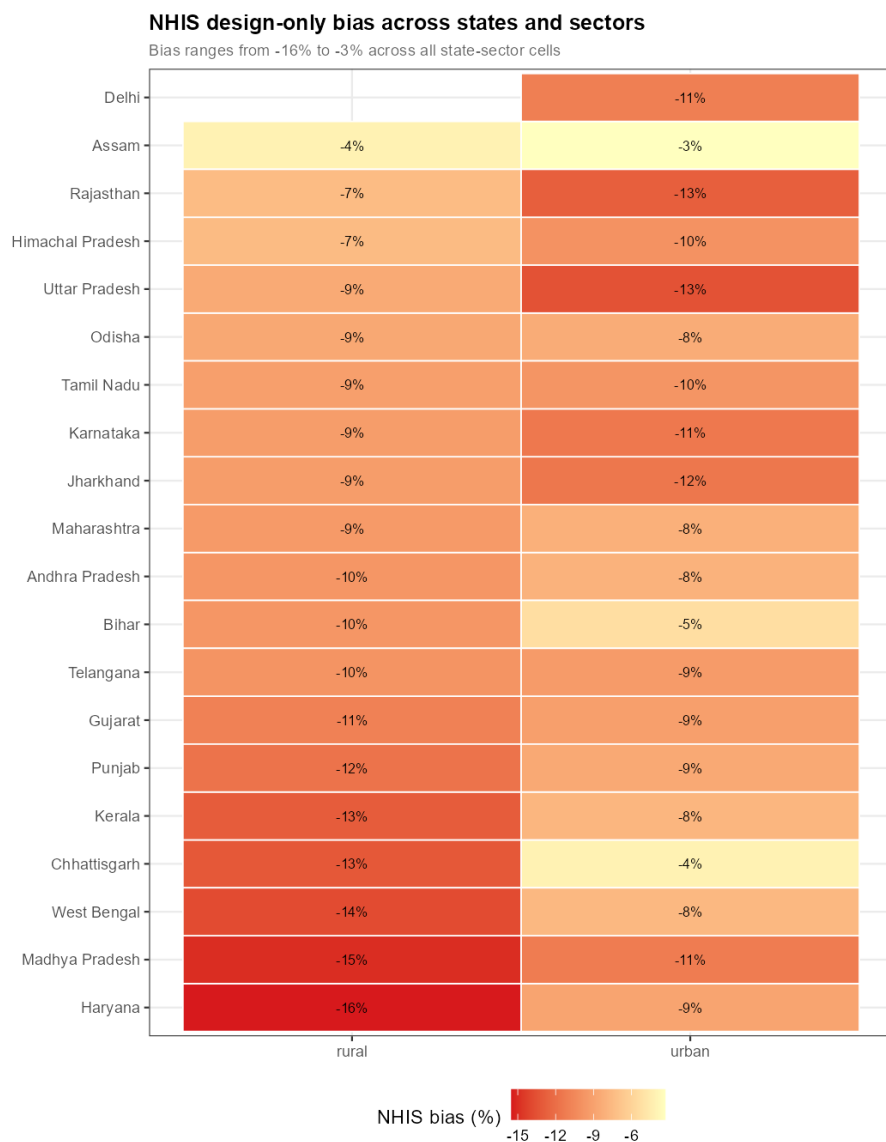


Panel-level design-only bias spans from 48.5 percent under-estimation in Panel 2, whose reference month is June, to 14.8 percent over-estimation in Panel 4, whose reference month is December. This range is large and reflects the structure of Indian seasonality. June is simultaneously the construction monsoon trough, the off-season for wedding-related food services, and a quiet month for retail and personal services. By contrast, December captures the wedding and festival commerce upswing across multiple industries. Panel 1 (March reference) shows mild under-estimation, and Panel 3 (September reference) shows mild over-estimation as the festival season begins.

The panel rotation design absorbs most of this variation. The all-India design-only bias of 9.8 percent is much smaller than the worst-panel bias of 48.5 percent, confirming that the rotation does what it is designed to do at the population level. However, panel averaging cannot reach zero because the Indian seasonal calendar is asymmetric. The June trough is deeper than the December peak when averaged across the industry mix, leaving a residual negative bias that no further rotation could absorb.

## State-level heterogeneity

**Figure 5: State-level heterogeneity in NHIS bias**



Design-only bias is negative across all 39 state-by-sector cells, ranging from approximately 3 percent to 16 percent under-estimation. The negative sign across every cell indicates that the issue is structural to the estimator rather than specific to any particular state's economy. The variation across cells reflects compositional differences in industry mix, drawn from PLFS 2025 Table 27, and the resulting exposure to seasonal troughs. States whose self-employment universe leans toward industries with sharper seasonal patterns face deeper bias. Any further industry-level decomposition of the state-by-sector gap would require analysis of PLFS unit-level microdata and is recommended for future work.

### *What this means for the national income distribution*

NHIS is being launched specifically to give India its first dedicated, large-sample dataset on the distribution of household income. Yet our analysis shows that even before reporting bias enters the picture, the Block 7c estimator under-measures non-agricultural self-employment income by close to 10 percent. With a modest reporting bias of 5 percent on revenue and 5 percent on expenses layered on, that figure approaches 40 percent. For a survey whose central purpose is to characterize the shape of Indian income, an under-measurement of this magnitude in a major segment of the working population is a material concern.

Statement 7 of the PLFS 2025 Annual Report indicates that own-account workers and unpaid family helpers together account for approximately 56.2 percent of India's working-age workforce (MoSPI, 2026). Block 7c is the single instrument that measures income for the non-agricultural slice of this group, around 14 to 15 percent of all Indian households (13 percent rural and 18 percent urban). The bias estimates in this paper apply to that slice alone. Even when they do not capture reporting biases from other blocks of the survey, such as the agricultural self-employment module, or capturing a higher bias than 5 percent on revenue and expenses, the under-measurement is substantial. The estimates documented here should therefore be read as a baseline floor for the under-measurement of incomes in this segment, with the actual divergence in the field likely to widen once agricultural and behavioural channels stack on top. Two implications follow directly.

First, the bias falls within a population whose household-level income has no equivalent measurement instrument. ASUSE 2022-23 collects detailed earnings data for unincorporated non-agricultural enterprises, but its unit of observation is the enterprise, not the household. ASUSE cannot tell us what share of household income comes from self-employment, nor how that household combines this source with other sources of income, and how earnings from a single enterprise might be shared across multiple households or among multiple members within the same household. NHIS, hence, is the only instrument designed to deliver household-level non-agricultural self-employment income that can be aggregated with the household's other income sources. The bias documented here therefore translates directly into the reported number for households that fall in this category.

Second, any practice that can be adopted to reduce reporting bias must be treated as non-negotiable in NHIS implementation. Our simulation shows that the design-only bias, before any field behaviour is introduced, is approximately 10 percent. The remaining 30 percentage points of under-measurement come entirely from the reporting layer. This means that the field implementation choices documented afterwards in this paper, including respondent selection, module sequencing, enumerator training, and visit timing, are not refinements at

the margin. They are the difference between a survey that delivers on its stated purpose and one that does not. We will elaborate on this in further sections.

### 3.1.2 Migration Linked Income

The current NHIS schedule does not include an explicit question identifying whether any household member is currently living elsewhere for work, or whether a former household member who has moved out still contributes financially to the household. Income earned by an absent member is captured only through the remittances module, which records money sent back to the household. Where a migrant member returns periodically but is not at home during the survey visit, income is recorded in the usual manner but through a proxy respondent. The former issue is addressed in this section while the latter shares its structure with the broader problem of proxy respondents answering in place of the primary earner and is hence addressed in further sections.

As with other predominantly agricultural economies, migration in India functions as a key strategy for households to escape stagnating or falling incomes. Having established earlier through Singh (2024) and interview evidence from Rohan Raj at IDinsight (2026), it is important to mention that migrant members of a household cannot be ignored in the design of an income survey.

A MoSPI field staff member who has been involved in data collection noted that this is less a coverage gap and more a mismeasurement problem. When the primary earner has moved out and is not treated as a usual household member, only remittances get captured, understating income. Even when the migrant is counted as part of the household, details about the income and through what sources it was earned remain unidentified (MoSPI Field Staff, interview, 2026).

The consequences of this gap extend beyond income measurement alone. De Brauw and Carletto (2012) argue that when household surveys do not identify migrant members, researchers are deprived of the ability to study how migration affects poverty and inequality at the household level. Without knowing which households have migrant members, it is not possible to trace how income composition changes after a member leaves, whether remittances are substituting for or supplementing other income sources, or whether migration is widening or narrowing gaps within the rural income distribution. For a survey whose core purpose is to measure inequality, this is a significant omission.

Other developing countries with large migrant populations have addressed this directly at the instrument level. Bangladesh's Household Income and Expenditure Survey includes a dedicated migration and remittance module in Section 8, which records extensive details for each migrant member. From when and where they migrated, their occupation, how frequently they send money to the method of transfer, and the value of goods sent home. The module explicitly flags migrant status before asking about remittances, ensuring that remittance income is recorded as a distinct and identifiable category rather than folded into

general transfer income. The Philippines takes a complementary approach at the survey architecture level. In addition to collecting remittance from family members in the Family Income and Expenditure Survey, they run a dedicated Survey on Overseas Filipinos as a rider to the quarterly Labour Force Survey, which distinguishes overseas contract workers from other migrant workers and tracks both the volume and method of remittances at the household level (Philippine Statistics Authority, 2024). While India's internal migration context differs from the Philippines' predominantly international one, better research requires remittances and migrant earnings to be separately identified to be accurately interpreted.

The recommendation that emerges is to add a short screening question to the NHIS household roster that flags whether any current or former household member is currently living elsewhere and contributes financially to the household. The screener could read: "Is there any current or former member of this household currently living elsewhere?" Including former members matters because in extended-family arrangements common in rural India, an adult child who has migrated for work may no longer be a usual member of the parental household but remains its principal source of income. For each absent member identified, the roster should record relationship to head, reason for absence, location, duration, remittance amount and frequency, and whether any income earned is otherwise reported in the household aggregate.

This is a low-cost modification as it only increases the duration of survey for a small yet important set of households. Even though it does not solve the deeper problem of capturing what and how the migrant earns, it does two things that matter. First, it makes migrant-sending households identifiable in the data, so that researchers can treat them separately in downstream analysis. Second, it ensures that remittance income is not recorded as generic transfer income but is associated with an identified absent earner, improving the interpretability of the income aggregate. Migrant-sending households often sit in the middle and upper-middle of the rural income distribution (Tumbe, 2014). Capturing migrant worker detail would ensure that such trends are better understood.

### 3.1.3 Imputed Rent

International standards, notably the Canberra Group Handbook on Household Income Statistics, recommend including the net value of housing services from owner-occupied dwellings in household income aggregates. India's current NHIS design excludes it. Both positions are defensible, and this section examines each before arriving at a recommendation.

The case for exclusion rests primarily on estimation difficulty. Imputing a rental value for an owner-occupied dwelling requires either observed rental market data or a reliable self-assessment from the homeowner. In rural India, rental markets are thin or effectively absent. Overall tenancy in India stands at around 20 percent of households, with states like Bihar

reporting rates as low as 2 percent (Himanshu, 2026). Where markets are this sparse, self-assessment methods produce highly variable values and introduce a new source of measurement error. Ceriani, Olivieri, and Ranzani (2019) document this problem directly, showing that self-assessed rental values by homeowners are frequently inflated due to sentimental attachment to the property, and that this inflation is difficult to detect or correct without external benchmarks. Himanshu (2026) further raises objection about how imputed rent is better understood as consumption from an asset than as current income. Including it in an income aggregate can raise measured income without reflecting actual purchasing power, which distorts both the level and the distribution of the income measure.

The case for inclusion rests on income distribution accuracy and international comparability. Wealthier households disproportionately own homes, and often more valuable ones. Excluding imputed rent therefore systematically compresses the upper tail of the measured income distribution, understating the gap between owners and renters and between wealthier and poorer households. Imputed rent also captures a component of economic wellbeing that is harder to conceal than cash income, which means its inclusion partially offsets the upper-tail compression introduced by underreporting of other income sources. Considering one of the major goals of the current income survey in India is to measure distribution better including the top percentile, inclusion of imputed rent makes more sense than not.

On comparability, countries that follow Canberra Group guidance cannot be compared directly to India's estimates, and part of the persistent gap between India's national accounts and survey-based income aggregates may reflect this omission. Where imputed rent is included, the Canberra framework requires deducting housing-related costs such as mortgage interest and regular maintenance to arrive at a net value, otherwise the imputation overstates the income flow for households still carrying a loan.

Bangladesh, Peru, and Indonesia all include imputed rent in their welfare aggregates using self-assessment methods (Ceriani, Olivieri, and Ranzani, 2019). The Bangladesh case is particularly relevant given that its rental market share of household consumption is comparably low, mirroring India's residential rental yields of 2 to 4 percent (Gandhi, Green, and Patranabis, 2022; Ceriani, Olivieri, and Ranzani, 2019). Ceriani et al. find that including imputed rent in Bangladesh produces statistically significant but modest changes in poverty and inequality, rarely exceeding one percentage point. More consequentially, rent imputation is not rank-preserving, meaning the composition of who counts as poor shifts even when aggregate indices do not. These findings reinforce both sides of the debate.

Finally, we believe that exclusion is defensible for rural India, where rental markets are too thin to support reliable imputation. For urban India, where rental markets are active and the

share of tenants is substantially higher, imputed rent should be included in the income aggregate. This distinction matters because it enables meaningful comparisons between renters and owners within urban areas, and across cities of different sizes, which the current design cannot support.

## 3.2 Implementation Issues

### 3.2.1 Timing of the Field Visit

The pre-testing report highlights two related findings. Income data is meaningfully more dependent on reaching the primary earning member of the household than consumption data is, because non-earning members typically have only partial knowledge of income sources. The time of day at which the enumerator arrives materially affects both the availability of the right respondent and the quality of responses given. Mornings tend to be rushed, with earners leaving for work, while evenings bring fatigue. In a survey that already averages 100 minutes of completion time, respondent patience is a binding constraint throughout (MoSPI, 2025).

This problem connects directly to the simulation findings. When reporting error is introduced at plausible levels into the Block 7c estimator, the resulting bias stacks substantially on top of the structural design bias already documented. The magnitude of that reporting-error layer depends directly on whether the respondent is the primary earner or a proxy. A proxy is both more likely to under-report out of uncertainty and less able to provide accurate expense figures. The reporting-bias channel and the field-timing channel are therefore not independent problems. Anything that increases the share of interviews conducted with the primary earner reduces the reporting-error layer compounding on top of the design bias.

Current MoSPI field practice, as described by a survey team member involved in CPI data collection and health modules under the HCES, follows standard office hours of approximately 9AM to 5PM with limited flexibility towards weekends (MoSPI field staff, interview, 2026). This may have been adequate for consumption surveys, where a non-earning household member can often provide sufficiently accurate responses. For an income survey, it is not. The pre-testing report itself acknowledges this, recommending that enumerators consider flexibility in survey scheduling to improve the chances of reaching the right respondent. The gap between that recommendation and current field practice is worth closing explicitly in the NHIS protocol.

IDinsight's income-focused survey work offers a more recent benchmark. In data collection conducted across states including Telangana, enumerators did not restrict themselves to a

single visit at a fixed time. Where the primary earner was unavailable, teams would call the household head later during the ongoing survey or return for a second or third visit when the member was present (R. Raj, interview, 2026). This multi-visit, flexible-timing approach increases both accuracy and cost. For an income survey where proxy reporting introduces a compounding layer of mismeasurement, that tradeoff is likely worth making. Rohan Raj further noted that early morning and evening visits, outside standard working hours, also yielded usable responses for income modules, suggesting that the time window for productive income interviews is broader than what current MoSPI recommendations in the pre-testing report assumes.

Occupation-specific scheduling offers further relief. Certain occupational groups have predictable off-days that do not fall on weekends. Casual labourers in some contexts are more reliably available mid-week, and targeting those households on those days meaningfully increases the probability of reaching the primary earner directly (R. Raj, interview, 2026). As the economy diversifies, this kind of scheduling intelligence becomes more rather than less important. Gig and platform workers, whose hours are irregular and who may work evenings or weekends, present a harder scheduling problem than any traditional occupation. Reaching them may require appointment-based visits or direct calls.

### 3.2.2 Gender Dynamics and Survey Sequencing

The pre-testing report recommends a gender-balanced enumeration team, and the reasoning connects directly to the timing issue discussed above. In a large-scale survey covering more than 450,000 households (GoIStats, Twitter, 2026), the ability to schedule repeated visits to reach the primary earner is limited. In practice, a substantial share of interviews will be conducted with whichever household member is available at the time of the visit, which during daytime hours is frequently a woman managing the household. Enumerator gender then becomes a material variable. In settings where female respondents are uncomfortable interacting with male enumerators, or where community norms restrict entry of male outsiders into the home, the interview either does not happen or produces lower-quality responses on sensitive questions (Fuletra and Joshi, 2024).

This problem is compounded by the nature of Indian households. As Dr. Deepak noted, houses are small and communities are tightly knit. When an enumeration team arrives, it is not just the household head who becomes involved. Neighbours, family members, and others may gather, increasing the time taken and making it harder to conduct a focused interview. Bangladesh's HIES deploys mandatory teams consisting of a senior officer, two enumerators, and two female facilitators for each household visit. The facilitators serve a dual function where they engage with female household members and also manage the

social dynamics around the interview, while the enumerator collects responses from the primary respondent. Dr. Deepak and Isha Fuletra both independently highlighted this kind of team structure as particularly relevant for India, where managing the social environment around the interview is as important as the interview itself (Deepak, interview, 2026; I. Fuletra, interview, 2026).

The need for female enumerators specifically is well established in the literature. Fuletra and Joshi (2024) find that gender matching between enumerators and respondents improves both participation rates and response quality, particularly for sensitive modules. The income survey context amplifies this. With 95 percent of pre-testing respondents reporting discomfort with income-related questions (MoSPI, 2025), the barrier to honest disclosure is already high. A female respondent interviewed by a male enumerator in a conservative household setting faces an additional barrier that has nothing to do with the question and everything to do with who is asking it.

The practical challenge is that women are significantly underrepresented among enumerators who pass MoSPI's current selection process, and in enumerator positions across organizations more generally. This was confirmed by staff at MoSPI, IDinsight, and by Sharad Pandey, Senior Researcher at ResGov, all of whom noted that the rigour of current entry requirements works against female recruitment. Enumerator work is time-consuming, requires interacting with multiple members across multiple households, and often involves travel through unfamiliar areas at unpredictable hours. Safety and mobility concerns therefore restrict the pool of women who would consider applying in the first place, regardless of whether they could meet the entry requirements (I. Fuletra, interview, 2026; Deepak, interview, 2026). Isha elaborated on a few recommendations that were followed at IDinsight. First, lowering entry requirements for female recruits widens the pool without compromising the overall quality of the team, since extended training can compensate for lower initial experience levels. Second, recruiting locally rather than centrally improves access and retention, since local familiarity reduces the mobility friction that deters non-local women from applying. Crucially, deploying female enumerators as part of a team rather than alone, along the lines of the Bangladesh HIES structure where two female facilitators accompany the enumeration team on each household visit, addresses the safety concern directly while also improving response quality on sensitive modules. In the MoSPI context, this could be operationalized by contracting female enumerators on a fixed-term basis specifically for the NHIS, ensuring that each field team includes at least one or two female members regardless of the composition of the permanent staff.

On survey sequencing, the issue it tries to address was identified at the instrument-level implication which follows from the respondent dynamics described above. Placing the short expenditure module (Block 10 of the NHIS 2026) before the income module could turn out

to be crucial and serves two purposes. First, it starts with questions that female respondents, who are often the ones available during daytime visits, can answer with confidence, building engagement before moving to the harder and more sensitive income questions. Second, it enables a live CAPI consistency check. If reported income falls below expenditure already recorded, the enumerator can probe gently before moving on. This check only works in one direction. If income is asked first, respondents have already anchored their expenditure answers to disclosed income, making any subsequent check far less meaningful.

Interview evidence from Rohan Raj and Sharad Pandey, both of whom have run similar surveys on the ground, supports the direction of this argument, though with caveats. Rohan Raj noted that in surveys combining asset and income modules (which generates a similar sequencing problem), his team consistently placed the asset questions first. The rationale was that assets are more concrete and verifiable, and asking income first risks an anchoring effect where respondents consciously or unconsciously adjust their subsequent answers to appear internally consistent with what they have already disclosed (R. Raj, interview, 2026). He raised two operational concerns too. First, real-time CAPI inconsistency flags, where expenditure exceeds income, can be difficult for enumerators to handle gracefully. Pushed to reconcile the two, enumerators may prompt respondents toward forced alignment rather than letting the inconsistency stand as a data quality signal. Second, and more delicately, where expenditure and income are answered by different household members, flagging an inconsistency mid-interview can create friction between them, with the female member who answered the expenditure questions honestly potentially bearing the consequences. He also raised a concern around the respondent availability and who answers the survey. In practice, the first point of contact during a survey visit is often a male household member, approached for consent and permission to proceed. Asking income questions while that member is present and engaged is operationally efficient. Moving expenditure to the front risks losing that window, since the male member may disengage or leave before the income module is reached, requiring a callback or revisit that adds cost and logistical complexity.

Sharad Pandey agreed on the potential value of this sequencing too but was careful to note that direct experimental evidence on this specific pair of modules is thin. Most ordering studies examine fatigue or within-module sequencing effects rather than the income-versus- short expenditure module pair specifically (S. Pandey, interview, 2026). Through his own fieldwork, he observed that when income is asked first and under-reported, expenditure answers that follow tend to drift downward to match, often not deliberately, and that a difficult income conversation leaves the respondent guarded for the remainder of the interview. This was in compliance to our hypothesis. He also then flagged that switching the module order creates a break in the NHIS series that data users are unlikely to welcome, and

that without proper retraining, enumerators tend to revert to the original flow regardless of what the instrument specifies. Some income sub-modules, particularly self-employment and agriculture, are also designed so that the income discussion primes expenditure recall, meaning the reorder requires rethinking enumerator training, not simply moving sections around.

Since the expenditure module collected in Block 10 of the current NHIS 2026 survey is short, fatigue before reaching the income modules is expected to be considerably lower than in a full expenditure survey, which softens the more operational of these concerns. The staggered or module-based approach, where different respondents answer different modules in a single visit, offers one way to address the availability concern while preserving the sequencing benefit.

The Philippines FIES captures expenditure before income, while Bangladesh's HIES does the opposite. The current NHIS places the consumption expenditure module toward the end of a 100-minute instrument, after income has already been asked. The concern about a series break, while valid, carries less weight here. The NHIS is in its first round and will inevitably undergo design changes as early fielding experience accumulates. Incorporating a sequencing change at this stage is considerably less disruptive than revisiting it after multiple rounds have established user expectations. Given this, we recommend piloting expenditure-before-income sequencing in a split sample in the next round, using a staggered module-based approach where operationally necessary, before considering a wider switch. Enumerator retraining would need to accompany any such pilot to prevent reversion to the original flow.

### 3.2.3 Enumerator Training

Enumerator training for income surveys requires a different standard than what currently exists. Sharad Pandey noted that enumerators during government programmes typically receive minimal training before deployment, and that what training exists tends to be theoretical rather than grounded in field realities (S. Pandey, interview, 2026). This was confirmed by a MoSPI survey team member who has been involved in data collection across modules (MoSPI field staff, interview, 2026). Income surveys present scenarios that simply do not arise in consumption surveys. These include managing refusal from higher-income households, handling respondent anxiety about tax implications, and recognizing when stated income is implausible given observed assets in the household. Moore, Stinson and Welniak (2000) document that social desirability bias is one of the most consequential sources of income measurement error in surveys, precisely because enumerators who are

not trained to handle sensitive income questions tend to accept implausible responses rather than probe gently. Without targeted training, these problems compound silently.

IDinsight's approach to training offers a concrete model. Rather than walking enumerators through abstract procedures, training was built around an extensive set of realistic situational scenarios covering the range of difficult field encounters (R. Raj, interview, 2026). Enumerators who had encountered a situation in training were more likely to respond appropriately in the field than those who had only read about it. Rohan Raj also mentioned how he trained his enumerators to ask contextually interesting questions of their own, outside the formal instrument, to keep respondents engaged through a long and demanding interview. These were not recorded and did not affect the data, but they served as natural breaks between tiring income modules, reducing fatigue and the satisficing behaviour that comes with it.

Finally, good training also ensures how enumerators frame the survey to respondents. Many households are deeply skeptical of income surveys, partly because of concerns that disclosure could expose them to scrutiny from tax or other government authorities. Reinforcing the strictly anonymous nature of the data, and clarifying that survey records are held within the statistical department and not shared with any other ministry or arm of government, is both factually accurate and important for building the trust needed to obtain honest responses (Moore, Stinson and Welniak, 2000). Alongside this, framing participation as showing patriotism and contributing to the state's ability to plan and allocate resources effectively speaks to a sense of civic purpose that resonates across income groups. The purpose could be repeated whenever it is felt that the respondents are getting nervous or reluctant to answer highlighting data's anonymousness.

### 3.2.4 Reaching the Top of the Income Distribution

Higher-income households are harder to access and more likely to refuse participation, and the pre-testing report documents both (MOSPI, 2025). This is not a problem that enumerator training or scheduling flexibility can meaningfully address. It reflects structural incentives. Participating in a government-led income survey offers these households no benefit while raising real concerns about scrutiny and tax implications. Gated communities frequently restrict enumerator entry altogether. Self-enumeration, sometimes proposed as a workaround, does not resolve the problem either. A household that will not open the door to an enumerator is unlikely to voluntarily complete a detailed income questionnaire for the government. No realistic adjustment to the field protocol will produce representative coverage at the top of the income distribution.

The more productive question is therefore not how to capture the top directly but how to make the survey data useful for characterising the top through other means. The established international approach is to integrate household survey data with administrative tax records, using the survey to anchor the middle of the distribution and tax data to characterise the upper tail. Atkinson, Piketty and Saez (2011) demonstrate that top income shares, which are highly sensitive to measurement at the upper tail, cannot be reliably estimated from survey data alone. For India specifically, Chancel and Piketty (2019) show this constraint directly. Their estimates of Indian income inequality from 1922 to 2015 rely on tax tabulations for the top of the distribution precisely because household surveys, which cover the bottom 90 percent of earners, cannot capture the top 10 percent with sufficient accuracy. With income tax data covering only around 7 percent of the Indian adult population at the time of their study, even this integration required significant methodological work.

Blanchet, Flores and Morgan (2022) formalize this approach into a replicable method. Their procedure identifies a merging point in the income distribution above which survey data becomes progressively less representative, and replaces it with reweighted tax data while preserving the microdata structure of the original survey. The tax data is also very helpful as, even where coverage is incomplete, it sets a credible lower bound on the number of people at given income levels. Applied to five countries including less-developed settings, the method consistently shifts measured inequality upward relative to survey-only estimates. For the NHIS, the practical implication is that the survey does not need to solve the top-income coverage problem itself. It needs to be designed so that such integration is feasible when the data is used downstream.

Making that integration feasible requires the survey to collect, for every respondent, a set of variables that can be matched or statistically linked to income tax records. Civil servants from India's economics, revenue, and audit services consulted for this paper agreed on the feasibility of this approach and offered specific guidance on what those variables need to be. Mapping their guidance against the draft schedule reveals that some variables are already captured adequately and could be used to conduct this, while some need further refinements:

- Usual activity status (Block 3, cols. 8–11): adequate as drafted, since the fifteen-category coding broadly corresponds to ITR filer categories.
- Secondary activity status (Block 3, cols. 9 and 11): adequate, and important to retain for members with multiple income sources.
- NIC code for non-agricultural activity (Q4.5, Q7.9): currently 2-digit; should be recorded at 3 or 4-digit level. The Income Tax Department uses its own CBDT-defined business code list, distinct from NIC, for ITR-3, ITR-4, ITR-5, and ITR-6 filings (Central

Board of Direct Taxes, 2024). A finer NIC code in NHIS is what makes a crosswalk between the two systems feasible.

- Legal form of enterprise: currently absent; a single categorical question (proprietorship, partnership, LLP, private limited, other) should be added to Block 7c. Legal form determines which ITR is filed, with sole proprietorships filing ITR-3 or ITR-4, partnerships and LLPs filing ITR-5, and companies filing ITR-6 (Income Tax Department, 2026). Without this variable, NHIS-recorded enterprises cannot be matched to the correct row of the tax data.
- District of residence (item 1.3): adequate as drafted.
- Highest educational level attained (Block 3, col. 7): adequate as drafted.
- Spousal labor market participation: recoverable from the existing household roster, no change needed.

This approach is not specific to high-capacity statistical systems. In Brazil, the federal revenue authority (Receita Federal) has released income tax tabulations from the DIRPF since 2015. Medeiros, Souza and Castro (2015) combined these with the PNAD household survey and found that survey-only estimates had overstated the decline in inequality during the 2000s. More recent work using linked administrative microdata places the top one per cent share at 27.4 per cent of national income in 2019, around seven percentage points higher than survey-only estimates (Palomo et al., 2025). South Africa has gone further institutionally. The South African Revenue Service shares anonymised personal income tax microdata with accredited researchers, which Chatterjee, Czajka and Gethin (2022) used alongside the National Income Dynamics Study to estimate that the top one per cent owns 55 per cent of household wealth. Malaysia is the more direct comparator for India. Personal income tax there covers around 18 per cent of adults and the Inland Revenue Board publishes only aggregated tabulations, not microdata. Khalid and Rosli (2025) applied the Blanchet, Flores and Morgan (2022) reweighting procedure to the Malaysian Household Income Survey using these tabulations and found the corrected Gini around two percentage points higher than the survey-only estimate. India's tax coverage is lower still, at roughly 8 to 9 per cent of adults in 2023-24 (Ministry of Finance, 2024), which strengthens rather than weakens the case for integration, since survey under-representation at the top is more severe relative to what tax data could correct. The integration approach is feasible at any level of fiscal capacity, provided the revenue authority releases tax data at sufficient granularity and the survey instrument is designed to support matching. India is well placed on the first condition, since the Income Tax Department already publishes Time Series Data and Income Tax Return Statistics. The binding constraint is the second.

A complementary source of triangulation was also identified in the interviews through the auditory services officer. India has state-level surveys and administrative datasets that record mean income levels for workers in specific occupations in the informal sector. These are not widely used for national inequality estimation but could serve as a plausibility check against NHIS-reported incomes for the same occupational groups in the same states. This consistency check is lower cost and more immediate than full tax-record integration, and it is particularly valuable for the informal sector where tax data has no coverage at all.

The conceptual distinction that frames this section is between capturing the mean of the income distribution and capturing its shape. A survey designed to capture the mean must include the top, because the top contributes disproportionately to it. A survey designed to capture the shape of the distribution can afford to miss the extreme top, provided it accurately captures the distribution below it and pairs with external data to characterise the upper tail. The NHIS should explicitly adopt the latter framing. Its primary obligation is to measure the income distribution across the bottom 90 to 95 percent of households with accuracy. Its secondary obligation is to be designed in a way that enables researchers to use tax data to extend that distribution upward. The recommendation that follows from this is twofold. First, the NHIS instrument should align its occupation, geographic, and ownership variables with the coding conventions used in income tax administration. Second, the government should expand the availability and frequency of income tax microdata releases, which currently remain aggregated and infrequent, to make survey-tax integration a practical rather than theoretical possibility for researchers working with the NHIS data.

## 4. Recommendations

The following recommendations emerge from the analysis. They are organized chronologically in the order of the preceding sections, covering survey design issues first and implementation issues second. Where recommendations require no change to the field protocol or instrument length, this is noted.

### Survey Design

1. The synthetic data exercise tells us that we should retain the last-month recall window in Block 7c (Q7.9 to Q7.12) for the current round of the NHIS. The simulation shows a design-only bias of around 10 per cent at the all-India level, which is admissible considering the trade-off between longer recall periods causing more misreporting. Misreporting hurts the data far more than the recall length itself, particularly given that the share of seasonal and sequential non-agricultural enterprises currently stands at only 1.5 per cent of non-agricultural self-employed households, consistent with the NSS 73rd Round finding that 98.3 per cent of

unincorporated non-agricultural enterprises are perennial (MoSPI, 2016). This small group does face severe under-measurement of 50 to 75 per cent, but at 1.5 per cent of the population the impact on national estimates remains limited. If the first round of NHIS finds the share of seasonal and sequential businesses materially higher than this, the trade-off would need to be recalibrated and a longer recall period (such as a three-month window or a peak-lean-reference combination) piloted in a subsequent round. This is unlikely to be needed.

2. The Pass B simulation from the synthetic data test tells us that issues pertaining to underreporting in Block 7c are far more expensive than they appear. A conservative 5 per cent revenue under-report and 5 per cent expense over-report shows up as a 30 percentage point underestimation of income for this segment, since the months-of-operation multiplier amplifies the field-level error by up to twelve. This 30 percentage point gap should be treated as the base case for reporting-driven under-measurement, which the field-implementation recommendations that follow are designed to address. Recommendations 8 through 10, on visit timing, enumerator training, and module sequencing, are therefore not refinements at the margin. They are the lever that determines whether Block 7c estimates are usable.
3. Add a single screening question to the NHIS household roster (Block 3) reading: "Is there any current or former member of this household currently living elsewhere?" For households that respond yes, a short follow-up sub-block should record, the relationship to head, reason for absence, current location, duration of absence and if they support the family financially. The remittances entry in the existing transfers module should then be linked to this absent member rather than recorded as generic transfer income. This is a low-cost modification because the screening question adds only one item to every roster, while the detailed sub-block is used only for the small but policy-relevant share of households with a migrant member, estimated at around 5 to 18 per cent depending on state. The full instrument length is therefore unaffected for the majority of the sample.
4. Ensure that operating costs recorded across the self-employment, agricultural income modules (Blocks 7a, 7b, 7c, Q7.2, Q7.7, Q7.10) as well as the expenditure module (Block 10) reflect actual prices paid after subsidies rather than market prices. The current instrument captures expenditure at the point of transaction, which in principle already incorporates subsidized input costs. This approach should be explicitly preserved in the NHIS design guidance and verified during pre-testing and enumerator training.

5. Include imputed rent in the income aggregate for urban households only. Add a question to Block 4 asking owner-occupier households in urban areas to estimate the monthly rent their dwelling would command if let on the open market, alongside the existing Q4.12 and Q4.13. The figure should be carried into the income aggregate net of housing-related costs, including mortgage interest already recorded in Q4.17 and routine maintenance recorded in Q10.20. For areas where rental markets are thin or for rural areas, the question could be left with a specific code highlighting the absence of rental market. This is in compliance with the Canberra Report and is essential to measure the distribution of income well.
6. Run an embedded split-sample experiment in the next round of the NHIS, allocating around 10 per cent of the sample (approximately 45,000 households) to a treatment arm with Block 10 moved to precede Blocks 5 through 9, and the remaining 90 per cent to the standard sequence. Treatment-arm enumerators should be retrained to prevent reversion. The two arms can then be compared on reported income, expenditure, and the income-to-expenditure ratio to estimate the sequence effect. This allocation is standard practice for embedded survey experiments at national statistical offices, including Statistics Netherlands in the Dutch Labour Force Survey (Van den Brakel, 2008), and is large enough to detect a meaningful national-level effect while keeping 90 per cent of the sample on the standard instrument so headline statistics remain on a stable basis. If the experiment shows that the new sequence improves response quality without distorting comparability, it can be adopted in the next round. A separate embedded experiment in a subsequent round can then test the live CAPI consistency check, layered on top of whichever sequence is adopted.
7. Refine three variables in the NHIS instrument so that the data can be statistically linked to income tax records by downstream researchers. First, the NIC code for non-agricultural activity in Q4.5 and Q7.9 should be recorded at 3 or 4-digit level rather than the current 2-digit level, to enable a crosswalk to the CBDT business code list used in ITR filings. Second, a categorical question on the legal form of enterprise (proprietorship, partnership, LLP, private limited, other) should be added to Block 7c immediately after Q7.9. Third, MoSPI and the Income Tax Department should agree upfront on a common occupation classification for the usual activity status in Block 3 cols. 8 and 10, so that the existing fifteen-category coding maps cleanly onto ITR filer categories. The remaining matching variables, including district of residence (item 1.3), educational attainment (Block 3 col. 7), and spousal labour market participation, are already captured adequately.

## Implementation

8. Revise the NHIS field protocol to extend the visit window beyond standard 9 AM to 5 PM hours. Early morning, evening, and weekend visits should be treated as standard, and at least one contact attempt per household should be scheduled outside usual working hours. Where the primary earner is unavailable during the in-person visit, the enumerator should collect what is possible from a proxy member and follow up with the primary earner by phone for the income blocks, sequencing the call so that the earner provides the income figures themselves rather than confirming what a proxy already said. Field plans should incorporate occupation-specific scheduling, targeting casual labourers mid-week and gig workers by appointment. Where proxy reporting on the income module is unavoidable, the response should be flagged in a new dataset variable.
9. Ensure each NHIS field team includes at least one female enumerator, structured along the lines of the Bangladesh HIES team, which deploys two female facilitators alongside enumerators. Three practical pathways are available. First, lower entry requirements for female recruits, in line with the NHIS pre-testing report's own recommendation on team composition (MOSPI, 2025). Second, extend the training period to compensate for lower initial experience. Third, recruit locally rather than centrally, since local familiarity improves both access and response quality, particularly in conservative settings.
10. Redesign enumerator training for the NHIS around realistic situational scenarios rather than abstract procedural guidance. Training should specifically cover managing refusal from higher-income households, handling respondent anxiety about tax implications, recognizing implausible income-to-asset ratios, and knowing when and how to probe gently without triggering disengagement. Enumerators should also be trained to maintain respondent engagement across a long interview through contextually appropriate and sometimes redundant yet entertaining conversation, and to open each interview by clearly communicating the anonymous and confidential nature of the data and its purpose in supporting national planning.

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# Appendix 1

## Derivation of Household Shares in Non-Agricultural Self-Employment

This appendix documents the derivation of the estimated share of Indian households engaged in non-agricultural self-employment operated through a proprietary or partnership enterprise, reported separately for rural and urban sectors. These estimates underpin the 13 percent rural and 18 percent urban figures used in the synthetic population construction. All source figures are drawn from the Periodic Labour Force Survey Annual Report, July 2024 to June 2025 (Ministry of Statistics and Programme Implementation, 2026).

### A1.1 Source figures

**Table A1.1.**

| Parameter                                                                | Value | Source (PLFS 2025)      |
|--------------------------------------------------------------------------|-------|-------------------------|
| Rural households: self-employed in non-agriculture                       | 16.4% | Table 3, all-India row  |
| Urban households: self-employed (combined, not sector-split)             | 31.7% | Table 4, all-India row  |
| Urban workers: share in agriculture (NIC Section A)                      | 7.5%  | Statement 8             |
| Urban agricultural workers who are self-employed                         | 74.1% | Table 20, urban persons |
| Urban workers: share self-employed (overall)                             | 40.4% | Statement 7             |
| Rural non-agricultural workers in proprietary or partnership enterprises | 79.2% | Statement 9, persons    |
| Urban non-agricultural workers in proprietary or partnership enterprises | 66.0% | Statement 9, persons    |

### *A1.2 Rural estimate*

The PLFS classifies rural households into six types, two of which isolate self-employment in agriculture and self-employment in non-agriculture separately. The rural share of households in non-agricultural self-employment can therefore be read directly from Table 3 as 16.4 percent. Multiplying this by the share of rural non-agricultural workers operating through a proprietary or partnership enterprise yields the share of rural households in this category:

**$16.4\% \times 79.2\% \approx 13.0\%$  of rural households.**

### *A1.3 Urban estimate*

The urban household typology in PLFS does not disaggregate self-employed households by sector. Table 4 reports only a combined self-employment figure of 31.7 percent. The non-agricultural share must therefore be imputed at the worker level.

The share of urban workers in agriculture is 7.5 percent, of whom 74.1 percent are self-employed. This gives  $7.5\% \times 74.1\% \approx 5.56\%$  of urban workers who are self-employed in agriculture. Since 40.4 percent of urban workers are self-employed overall, the non-agricultural share of urban self-employed workers is:

$(40.4\% - 5.56\%) / 40.4\% \approx 86.2\%$ .

Applying this non-agricultural share to the combined urban household figure, and then the urban proprietary or partnership share of 66.0 percent, yields:

**$31.7\% \times 86.2\% \times 66.0\% \approx 18.0\%$  of urban households.**

### *A1.4 Limitations*

Three approximations are compounded in these estimates. First, the non-agricultural share of urban self-employed households is imputed from worker-level data because PLFS does not publish an urban household-type breakdown by sector. Second, the proprietary or partnership share, measured among all non-agricultural workers, is applied to a household-level self-employment figure under the assumption that the ratio transfers between units of analysis. Third, because self-employed workers are concentrated in proprietary or partnership enterprises to a greater extent than non-agricultural workers on average, the resulting estimates are likely to be conservative lower bounds rather than point estimates. The 13 and 18 percent figures used in the paper should be read with these limitations in mind.

## Appendix 2

### Simulation design decisions

This appendix documents the parameter choices and design decisions underlying the synthetic-data simulation. Each table below makes explicit what was set, what data underlies the choice, and where stylisation was necessary in the absence of published data at the required granularity.

#### *A2.1 Industry mapping to NIC codes*

The simulation operates on eleven industry categories. Each category maps to a set of two-digit NIC-2008 codes and connects back to the relevant PLFS Table 27 NIC sections used for state-level industry shares.

**Table A2.1.**

| Industry category | Description                                                                | NIC-2008 codes (2-digit)           | PLFS Table 27 section |
|-------------------|----------------------------------------------------------------------------|------------------------------------|-----------------------|
| retail            | Retail trade                                                               | 45, 46, 47                         | G                     |
| food_svc          | Food services and accommodation                                            | 55, 56                             | I                     |
| personal_svc      | Personal services (repair, beauty, laundry)                                | 95, 96                             | S (subset)            |
| mfg_light         | Light manufacturing (food, textiles, apparel, leather, wood, paper)        | 10, 13, 14, 15, 16, 17, 18         | C (subset)            |
| mfg_heavy         | Heavy manufacturing (rubber, metals, machinery, motor vehicles, furniture) | 22, 23, 24, 25, 27, 28, 29, 31, 32 | C (subset)            |
| construction      | Construction                                                               | 41, 42, 43                         | F                     |
| transport         | Transport and storage                                                      | 49, 50, 52, 53                     | H                     |
| professional      | Professional services (legal, accounting, consulting, IT, R&D)             | 62, 69, 70, 71, 73, 74             | M                     |

|           |                                                             |                        |                      |
|-----------|-------------------------------------------------------------|------------------------|----------------------|
| health    | Health services                                             | 86, 87                 | Q                    |
| education | Education (private tutors, coaching, small private schools) | 85                     | P                    |
| other_svc | Other services (membership organisations, residual)         | 58, 61, 77, 81, 82, 93 | S (subset), residual |

### *A2.2 Sub-section splits within Section C and Section S*

PLFS Table 27 reports state-level industry distributions at the section level only. Two of our categories (manufacturing and personal services) require splits within a single section. These splits are derived from PLFS Table 26, which provides 2-digit NIC division shares at the all-India level. The splits are applied uniformly across states because PLFS Table 26 does not publish state-level data at this granularity.

**Table A2.2.**

| Section           | Split applied                   | Source                                                                                                                                                   |
|-------------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| C: Manufacturing  | 58% light, 42% heavy            | Derived from PLFS Table 26 by summing rural+urban person shares of light NICs (10, 13–18) and heavy NICs (22–32).                                        |
| S: Other services | 82% personal_svc, 18% other_svc | Derived from PLFS Table 26: NIC 95+96 (repair, hairdressing, laundry) account for ~82% of Section S workers; NIC 94 (membership organisations) for ~18%. |

### *A2.3 Monthly revenue seasonality multipliers*

Each industry is assigned a 12-month vector of revenue multipliers, expressed relative to the industry's annual mean of 1.00. The simulation uses the same multipliers across all states, since PLFS does not publish state-level seasonal income data at the granularity required. Expense seasonality is modelled as a damped (60% of revenue amplitude) and one-month-lagged version of the revenue pattern.

**Table A2.3.** Monthly revenue multipliers by industry, March through February. Values express revenue relative to each industry's annual mean.

| Industry     | Mar  | Apr  | May  | Jun  | Jul  | Aug  | Sep  | Oct  | Nov  | Dec  | Jan  | Feb  |
|--------------|------|------|------|------|------|------|------|------|------|------|------|------|
| retail       | 0.78 | 0.85 | 0.92 | 0.95 | 0.98 | 1.05 | 1.12 | 1.30 | 1.35 | 1.15 | 0.95 | 0.80 |
| food_svc     | 0.95 | 0.85 | 0.80 | 0.70 | 0.75 | 0.85 | 0.95 | 1.15 | 1.30 | 1.25 | 1.15 | 1.10 |
| personal_svc | 0.80 | 0.80 | 0.85 | 0.70 | 0.75 | 0.90 | 1.00 | 1.15 | 1.40 | 1.40 | 1.20 | 0.85 |
| mfg_light    | 0.85 | 0.90 | 0.95 | 1.00 | 1.00 | 1.05 | 1.20 | 1.15 | 1.10 | 1.00 | 0.95 | 0.85 |
| mfg_heavy    | 0.98 | 1.00 | 1.02 | 1.02 | 1.00 | 0.98 | 1.00 | 1.02 | 1.05 | 1.00 | 0.96 | 0.97 |
| construction | 1.25 | 1.15 | 1.00 | 0.65 | 0.55 | 0.50 | 0.65 | 0.85 | 1.15 | 1.30 | 1.25 | 1.20 |
| transport    | 0.95 | 1.05 | 1.10 | 0.85 | 0.80 | 0.85 | 0.95 | 1.10 | 1.15 | 1.10 | 1.10 | 1.00 |
| professional | 1.00 | 1.02 | 1.05 | 1.00 | 0.95 | 0.98 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| health       | 0.95 | 0.85 | 0.85 | 1.10 | 1.25 | 1.25 | 1.05 | 0.90 | 0.95 | 1.10 | 1.15 | 0.95 |
| education    | 1.10 | 1.25 | 1.20 | 0.70 | 0.65 | 0.85 | 1.10 | 1.20 | 0.90 | 0.80 | 0.75 | 0.80 |
| other_svc    | 0.95 | 0.95 | 1.00 | 0.95 | 0.95 | 1.00 | 1.05 | 1.10 | 1.10 | 1.05 | 0.95 | 0.95 |

#### *A2.4 Justification for each industry's seasonal pattern*

The numerical multipliers are stylised approximations based on widely understood Indian industry seasonality. These patterns are not estimated from a single empirical source. A deeper calibration using PLFS unit-level microdata, which captures earnings across all calendar months through its continuous fieldwork design, is recommended for future work.

**Table A2.4.**

| Industry     | Pattern justification                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| retail       | Diwali peak (October–November) drives festival shopping. Trough in March (post-festival lull).                                                                             |
| food_svc     | Wedding season (October–February) drives demand. Trough during monsoon (June–August).                                                                                      |
| personal_svc | Sharp Diwali and wedding-season peak (October–December) for beauty, tailoring, repair. Monsoon trough.                                                                     |
| mfg_light    | Modest seasonal pattern; peak during pre-festival inventory build-up (September–October).                                                                                  |
| mfg_heavy    | Essentially flat across the year; output less weather-dependent.                                                                                                           |
| construction | Sharp monsoon trough (June–August, falling to 50% of annual mean in August). Peak in dry pre-monsoon and post-monsoon months (March, October–December).                    |
| transport    | Modest peak around festival commerce months; mild monsoon dip.                                                                                                             |
| professional | Essentially flat across the year; activity not strongly weather- or festival-dependent.                                                                                    |
| health       | Counter-cyclical peak during monsoon (waterborne and respiratory illness, June–August) and winter (December–January). Trough during dry pre-monsoon months.                |
| education    | Sharp trough during summer school holidays (June–July) and post-board-exam period (January–February). Peak during exam preparation periods (April–May, September–October). |
| other_svc    | Modest pattern; mild peaks around festival commerce months.                                                                                                                |

## A2.5 Other simulation parameters

**Table A2.5.**

| Parameter                                | Value                                  | Justification                                                                                                                                                                                                                                          |
|------------------------------------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Households per business count            | 1: 0.91, 2: 0.08, 3: 0.01              | Consistent with ASUSE 2022-23 finding that 86.5% of unincorporated non-agricultural enterprises are sole own-account enterprises (MoSPI, 2024; Dvara Research, 2026).                                                                                  |
| Cat 3 prevalence                         | 1.0% of HH                             | Conditional probability 1.1% given single-business HH; consistent with NSS 73rd Round share of seasonal enterprises (MoSPI, 2016).                                                                                                                     |
| Cat 5 prevalence                         | 0.5% of HH                             | Conditional probability 5.6% given multi-business HH; bracketed by NSS seasonal share (floor) and IHDS multi-activity rate (ceiling).                                                                                                                  |
| Cat 3 operating window                   | 5–8 contiguous months                  | Reflects typical agricultural off-season trading, monsoon-affected construction, and school-year tutoring patterns described qualitatively in Harriss-White and Gooptu (2001).                                                                         |
| Cat 5 window length and gap              | 3–5 months active, $\geq 2$ months gap | Captures sequential seasonal activity patterns characteristic of India's unorganised economy (Harriss-White and Gooptu, 2001), where vendors and small operators shift across festival cycles, agricultural seasons, and weather-driven demand cycles. |
| Closure rate (months/year, perennial HH) | 0.2–2.0 (industry-specific)            | Highest for construction (2.0) and education (1.2) reflect structural shutdown periods. Stylized values were used to reflect industry-typical patterns.                                                                                                |
| Gross margin range                       | 8%–80% (industry-specific)             | Calibrated to typical thin-margin retail (8–20%), thicker-margin services (40–80%). Stylized values were used to reflect reported industry norms.                                                                                                      |
| Pareto threshold                         | Rs. 5,00,000 / month                   | Conservative threshold much bigger than the literature anchor identified by Chancel and Piketty                                                                                                                                                        |

|                                       |                                                  |                                                                                                                                                            |
|---------------------------------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       |                                                  | (2019) as the 94th-percentile floor of the Indian income distribution.                                                                                     |
| Pareto shape parameter (alpha)        | 2.2                                              | Mid-range value consistent with right-skewed informal income distributions (Aitchison & Brown, 1957; Chancel & Piketty, 2019).                             |
| Lognormal sigma (urban / rural)       | 1.05 / 0.95                                      | Calibrated to reproduce the within-state variance of self-employment earnings observed in PLFS Table 40, with rural distributions modestly less dispersed. |
| Reporting bias (Pass B)               | Revenue $\times$ 0.95,<br>Expenses $\times$ 1.05 | Conservative bounds assumed consistent with patterns documented in the NHIS pre-test report (MoSPI, 2025).                                                 |
| Within-household business correlation | 0.05 unrelated, 0.10–0.70 related                | Stylised values reflecting the assumption that related businesses (e.g. retail and food services) co-move more strongly than unrelated pairs.              |

## A2.6 Summary

The table below makes explicit which decisions are anchored in PLFS published data and which represent stylized approximations. Stylized choices are documented transparently rather than presented as data derived.

**Table A2.6.**

| Decision                              | Data available from PLFS?    | What we did                                 |
|---------------------------------------|------------------------------|---------------------------------------------|
| State-level industry distribution     | Yes (Table 27, worker-level) | Used directly                               |
| State-level mean earnings             | Yes (Table 40)               | Used as calibration target                  |
| Sector-level prevalence (rural/urban) | Yes (Tables 3 and 4)         | Used directly with derivation in Appendix 1 |
| Manufacturing light/heavy split       | Yes (Table 26, 2-digit NIC)  | Derived as 58/42 from worker shares         |

|                                                                          |                                              |                                                                                                         |
|--------------------------------------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Services personal/other split                                            | Yes (Table 26, 2-digit NIC)                  | Derived as 82/18 from worker shares                                                                     |
| Enterprise-level industry mix for Self-Employed Non-Agriculture Category | No (PLFS reports broader worker shares only) | Used worker shares as approximation; documented as limitation                                           |
| Monthly seasonality by industry                                          | No (not published at this granularity)       | Stylized approximations reflecting industry standards; PLFS microdata identified for future calibration |
| Closure rates and seasonal closure weights                               | No                                           | Stylized values reflecting industry-typical patterns                                                    |
| Gross margin ranges                                                      | Indirect (ASUSE 2022-23 enterprise-level)    | Stylized ranges within published norms                                                                  |
| Reporting bias rates                                                     | Indirect (NHIS pre-test report)              | Conservative 5%/5% applied multiplicatively                                                             |

## Appendix 3

### Robustness to alternative random seeds

This appendix reports the simulator's headline results under three alternative random seeds, in addition to the primary seed (20260419) used in the body of the paper. The purpose is to confirm that the bias estimates reported are stable to stochastic variation and that the choice of 75,000 households is sufficient for the precision claimed.

Each run uses identical parameters except for the random seed governing all stochastic draws. The resulting datasets differ only in the specific industry assignments, monthly revenue draws, panel allocations, and closure realisations for individual households.

**Table A3.1.** Headline bias estimates under four random seeds. Differences across seeds remain within 0.4 percentage points for every metric, confirming that 75,000 households provides adequate precision for the bias estimates reported. A is design-only bias, B is design plus reporting bias.

| Seed                      | All-India A   | All-India B   | Rural A       | Urban A       | Panel 2 A     |
|---------------------------|---------------|---------------|---------------|---------------|---------------|
| 20260419 (primary)        | −9.8%         | −39.6%        | −10.1%        | −9.7%         | −48.8%        |
| 20250101 (alt 1)          | −9.7%         | −39.4%        | −10.0%        | −9.6%         | −48.6%        |
| 20250715 (alt 2)          | −9.9%         | −39.7%        | −10.2%        | −9.8%         | −49.0%        |
| 20251231 (alt 3)          | −9.8%         | −39.5%        | −10.1%        | −9.7%         | −48.7%        |
| <b>Range across seeds</b> | <b>0.2 pp</b> | <b>0.3 pp</b> | <b>0.2 pp</b> | <b>0.2 pp</b> | <b>0.4 pp</b> |

*Across all metrics tested, the maximum range across the four seeds is 0.4 percentage points (Panel 2 design-only bias), with most metrics varying by 0.2 to 0.3 percentage points. This is well below the magnitudes of the substantive findings (a roughly 30 percentage point gap between Pass A and Pass B at the all-India level, and biases of 50 to 87 percent for partial-year and sequential households). The simulation is therefore stable under reasonable seed variation, and a sample size of 75,000 households is sufficient to support the precision of the headline estimates.*

## Appendix 4

### Reproducibility of the Synthetic Data Analysis

The synthetic data exercise described in Section 3.1.1 and Appendices 1, 2, and 3 is fully reproducible. All code and inputs required to regenerate the household-level dataset, the simulation outputs, and every figure and table in Section 3.1.1 are provided as a self-contained R project.

The project contains five R scripts: 01\_params.R defines all simulation parameters, 02\_generate.R builds the synthetic ground-truth population, 03\_simulate.R applies the NHIS Block 7c measurement rule under both passes, 04\_figures.R produces Figures 1 through 5, and run\_all.R orchestrates the pipeline. The simulation requires one external input, state\_level\_inputs.csv, which is a hand-compiled extract of state-level statistics from the PLFS 2024-25 Annual Report. Each column in this file maps directly to a published PLFS table, and the mapping is documented in the README that accompanies the code.

All stochastic choices in the simulation are governed by a single random seed (MASTER\_SEED = 20260419), so the headline bias estimates reported in this paper are bit-for-bit reproducible across runs on the same R version. Required R packages are data.table, ggplot2, and scales. Total runtime is approximately five to eight minutes on a standard laptop.

Running the pipeline produces three CSV files in the outputs directory that correspond to Tables 3, 4, and 5 in this paper, and five PNG files that correspond to Figures 1 through 5. Intermediate datasets, including the household ledger, business-level reference-month observations, and the state-by-sector bias summary, are written to a separate data directory and are available for any reader who wishes to verify, replicate, or stress-test the results.

## Appendix 5

### Brief Report on Pre-testing of Schedule of Household Income Survey

The pre-testing report on the NHIS 2026 schedule, prepared by the Ministry of Statistics and Programme Implementation, is attached on the following pages. The report documents the field-level findings from the schedule's pre-test exercise and is referenced throughout Section 1.3 and Section 3 of this paper, particularly in connection with respondent discomfort with income disclosure, reference-period comprehension, average interview duration, and access barriers among higher-income households. The report is reproduced here in full to support direct verification of the figures and observations cited in this paper.